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Chosen Unemployment or Chosen Career? : A Public Policy Evaluation of Separations by Mutual Agreement in France

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NON-CONFIDENTIAL REPORT

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Declaration of Academic Integrity

Hereby I, Salma EL AAZDOUDI, confirm that:

1. The results presented in this report are my own work.
2. I am the author of this report.
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Salma EL AAZDOUDI, 12 October 2026

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Abstract

Since 2008, French workers can end an open-ended contract by mutual agreement with their employer and still receive unemployment insurance. The reform was a wager: a higher cost for the insurance system in exchange for safer career transitions. In 2026, the government cut the benefit duration of these separations by mutual agreement (SMAs), on the premise that signatories chose to leave and therefore draw more heavily on benefits. This paper tests that premise with MiDAS administrative data, by comparing SMA signatories with workers who left an open-ended contract through another route and who are matched on 217 pre-separation characteristics.

Once selection is accounted for, the SMA only delays the return to wage employment slightly, and the gap closes after about two and a half years. This delay is not idle time: signatories use it to start a business, to train or to wait for a durable job, and they return at a similar wage. The cost for unemployment insurance reflects this: signatories receive slightly more benefits during the first two years, but fewer than comparable workers after six years. A sequence analysis of monthly trajectories explains this reversal. Most signatories follow the same path as their controls, a fast return to wage employment. The differences lie in two minority paths: signatories more often move into self-employment, although their profile gave them good wage employment prospects, while their controls more often cycle between short jobs and benefit receipt, which is what makes them costlier in the long run. Online forum posts confirm that, for many workers, the alternative to an SMA is not an insured exit but an uninsured resignation, or no exit at all. Rather than financing longer inactivity, unemployment insurance appears to finance chosen transitions, at no extra cost over six years.

Keywords: unemployment insurance, separation by mutual agreement, propensity score matching, event study, sequence analysis

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1 Introduction

"Even if that were the case, the agreement would be beneficial." France's then Minister of Labour, Xavier Bertrand, offered this reply to a parliamentary committee on 2 April 2008 when warned that a new labour reform might drain the unemployment insurance system [1]. He was defending a recent pact between employers' organisations and trade unions that introduced separations by mutual agreement (SMAs, known in French as *ruptures conventionnelles individuelles*) into French labour law. By enabling the mutually agreed termination of open-ended contracts (OECs, known in French as CDI, *contrats à durée indéterminée*)—and granting departing workers access to unemployment benefits—the reform raised an immediate financial concern as it threatened to channel an unprecedented number of workers onto the benefit rolls.

Yet Bertrand's reply is striking for what it concedes. He did not dispute the potential for increased claims. Instead, he framed this financial risk as the necessary price for an "unprecedented change of mindsets and behaviours" in the French labor market. Under the banner of flexicurity, the scheme promised a dual advantage. For employers, it offered flexibility and reduced the threat of litigation, aiming to narrow the divide between temporary contracts and OECs—the latter being historically rigid and costly to terminate. For workers, the goal was to secure career transitions and facilitate mobility. In Bertrand's words, the reform would "guarantee employees the flexibility they need to manage their career paths over their entire working lives."

On volumes, the warning proved accurate. The SMA has become a routine way of ending an OEC: some 515,000 were signed in 2024, between 8% and 21% of all OEC terminations depending on the sector. About three-quarters of them lead to a benefit claim, and these claims absorb 26% of unemployment benefit spending, or €9.4 billion (Unédic, 2026).

Eighteen years later, the government's stance has shifted, and the price Bertrand accepted is no longer deemed sustainable. This policy reversal occurred in two stages. Initially, from 2013, successive governments targeted employers, raising levies on SMA severance pay to deter firms from using the scheme as a disguised early-retirement route. By 2025, however, the critique turned explicitly toward the workers themselves. Arguing that the SMA is a "choice" exposed to "abuse," Labour Minister Astrid Panosyan-Bouvet demanded at least €400 million in annual savings from these claims. Consequently, the employer contribution rose from 30% to 40% in January 2026, and since September 2026, the maximum benefit duration for SMA claimants under 55 has been cut to 15 months—against 18 months for any other separation.

The 2008 reform was a wager: the added cost to unemployment insurance would buy better career transitions. The 2026 retrenchment assumes this wager was lost—that SMA signatories draw more heavily on benefits simply because they chose to leave. This paper tests that assumption. **Has the initial promise been kept, or only the cost incurred?**

The next section contextualises the SMA within the French labour market, reviews the established literature, and formally states the public policy problem the paper addresses and highlights

[1] Hearing before the Social Affairs Committee of the National Assembly, 2 April 2008, reported in Dord (2008). The committee record summarises the minister's words in reported speech; the translation is the author's. The full sentence reads: "Even if that were the case, the agreement would be beneficial, because moving towards a negotiated separation brings about an unprecedented change in mindsets and behaviour."

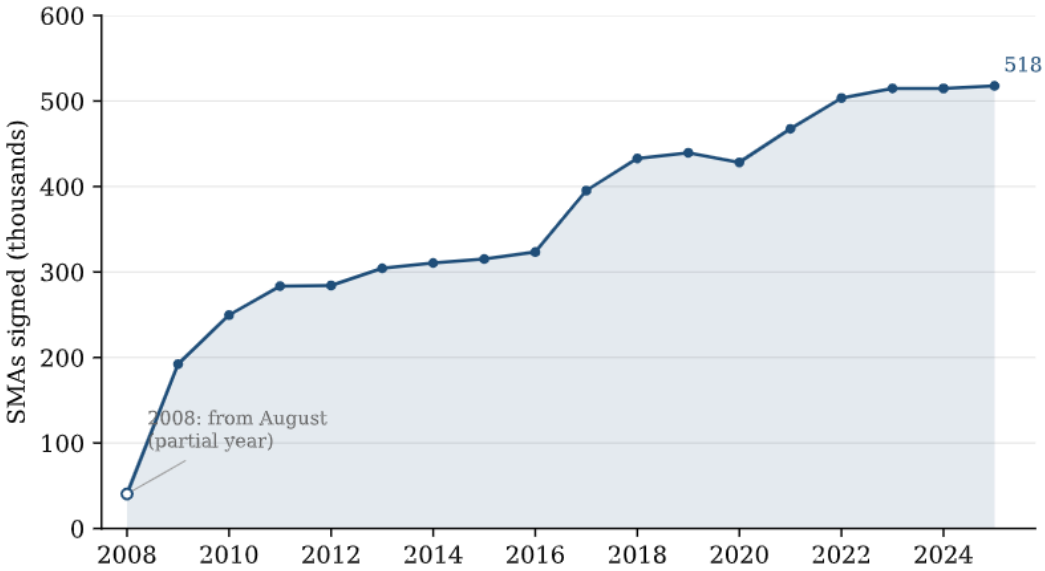


Figure 1: Aggregate Time Series : Number of SMAs signed between 2008 and 2025

Source: DARES, open data : mouvements de main d'oeuvre.

its contributions.

1.1 Contextualization

OECs account for approximately 72.9% of wage employment in France (Insee, 2025). Traditionally, these contracts carry robust employment protection and can only be terminated under restrictive conditions. Prior to the 2008 reform, France ranked eighth among OECD countries regarding the stringency of its regulations on individual dismissals. This stringency is the result of highly litigious legal framework. At the time the SMA was introduced, workers benefited from a 30-year statute of limitations to contest dismissals for personal reasons, and a 12-month window to challenge economic dismissals. Furthermore, the *Cour de cassation* (the French Supreme Court) applied a notoriously strict interpretation of what constituted valid grounds for dismissal (Andréo et al., 2026). .

To address this, Law No. 2008-596 "on the modernisation of the labour market" (*portant modernisation du marché du travail*) was enacted. Its rationale was to make OECs more flexible by allowing them to be terminated through a simplified procedure, while guaranteeing financial security for departing workers. Specifically, the legislation pursued three main objectives. Specifically, the legislation pursued three primary objectives (Berta et al.,2010):

- **Reducing segmentation:** Narrow the divide in the labour market between highly protected permanent contracts and temporary jobs.
- **Defusing separations:** Drastically reduce the volume of costly and lengthy disputes brought before the labour courts by providing a legally secure, mutually agreed exit route.
- **Securing mobility:** Facilitate job-to-job transitions by maintaining a worker's eligibility for unemployment insurance.



Figure 1: The separation by mutual agreement procedure

Notes: Stylised timeline of an individual separation by mutual agreement for an employee on an open-ended contract. Dark figure: employer; light figure: employee. The contract can end at the earliest the day after the approval. *Source:* French Labour Code, articles L.1237-11 to L.1237-16.

Figure 2: The separation by mutual agreement procedure

Notes: Stylised timeline of an individual separation by mutual agreement for an employee on an open-ended contract. Dark figure: employer; light figure: employee. The contract can end at the earliest the day after the approval. *Source:* French Labour Code, articles L.1237-11 to L.1237-16.

Eighteen years later, the rigidity that motivated the SMA has not disappeared. In 2025, France ranked seventh out of 36 OECD countries for the strictness of dismissal rules on open-ended contracts (Figure 3), and about 25% of dismissals for personal reasons are contested in labour courts, compared with 0.2% of SMAs Carry et al. (2025). The SMA offers a way around this: legal certainty for the employer, through administrative approval, and economic security for the worker, through unemployment insurance.

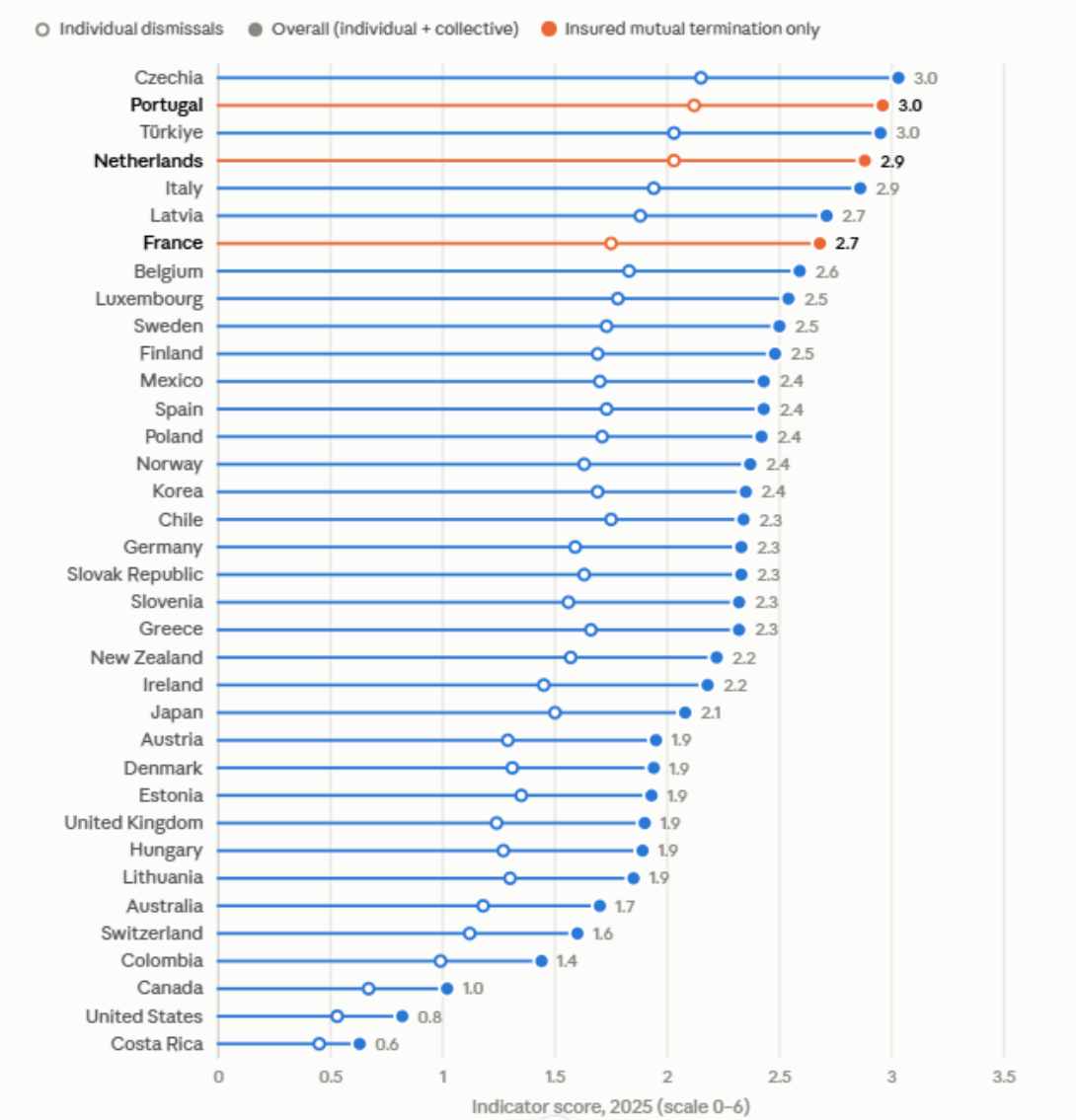


Figure 3: Overall strictness of regulation of dismissing workers on open-ended contracts, 2025

Notes: OECD employment protection indicators, version 4, regulation in force on 1 January 2025. Scores range from 0 to 6; higher values indicate more stringent dismissal regulation. *Source:* OECD Employment Outlook 2026, Chapter 6.

This pattern is not specific to France (Table 1). Only France, the Netherlands and Portugal give full access to unemployment insurance after a mutually agreed termination, and all three have comparatively strict dismissal rules. Insured mutual termination thus appears where dismissal is most constrained.

Table 1: Access to unemployment insurance after resignation and termination by mutual agreement

Regime	Country	Conditions after mutual termination
Mutual termination only, no penalty	France	All SMAs; maximum duration being cut from 18 to 15 months
	Netherlands	Only if the agreement states the employer's initiative
	Portugal	Only for restructuring or economic difficulties
Resignation and mutual termination, no penalty	Hungary, Slovak Republic	Same access as after dismissal
Resignation and mutual termination, with penalty	Germany	Waiting period, typically up to 12 weeks
	Switzerland	Suspension, often up to 2 months
	Denmark	Suspension, generally 3 weeks
	Sweden	Waiting period, most often 9–10 weeks
	Poland	Waiting period of up to 90–180 days if worker-initiated
	Estonia	Treated as voluntary quit: delay of several weeks or months
	Austria, United Kingdom	Sanction (in Austria, mainly after resignation)
No entitlement in either case	Canada, Spain, Greece, Italy, Luxembourg, Türkiye	–

Notes: Penalties apply at the start of the benefit spell. *Sources:* DG Trésor, *Séminaire Politiques de l'emploi*, 17 March 2026; OECD Employment Outlook 2026, Chapter 6.

1.2 Literature review

Whether the SMA has delivered on the three objectives it was assigned to is an empirical question, and the economic literature has examined them very unevenly. Table 2 summarises the studies that deal directly with the SMA and indicates which of the three objectives each one addresses.

Reducing segmentation To our knowledge, no study evaluates the effect of the SMA on the divide between permanent and temporary employment. What has been established concerns flows out of OECs rather than duality. Exploiting the staggered adoption of the scheme across establishments, Batut and Maurin (2019) find that separations from OECs rise by about 18–20% after first use, with no detectable effect on employment levels. Using a matched difference-in-differences design on firm data for 2006–2009, Signoretto (2016) likewise finds that user firms record more exits and more job destruction. The SMA thus increased labour market fluidity; whether it narrowed segmentation remains untested.

Table 2: Economic literature on the SMA and the three objectives of the 2008 reform

Study	Data and method	Main finding	Obj. 1	Obj. 2	Obj. 3
Panel A. Causal or quasi-causal evaluations					
Batut and Maurin (2019)	Establishment data; event study around first use	OEC separations rise by 18–20%; about 10% of personal dismissals replaced; no employment effect	○	●	–
Signoretto (2016)	Firm data, 2006–2009; matched difference-in-differences	More exits and job destruction in user firms; weak substitution for dismissals	○	●	–
Milin (2018)	Labour turnover data, 1993–2017; trend counterfactual	About 75% of SMAs replace resignations; no significant substitution for personal dismissals	–	●	–
Carry and Schoefer (2024)	Aggregate and cell-level series, court statistics, HR survey	About 12% of personal dismissals converted; labour-court cases after dismissal do not decline	–	●	○
Panel B. Descriptive evidence					
Bourieau (2013)	Survey of 4,502 signatories, 2011	48% report a joint decision, 29% felt compelled to leave; 69% cite access to benefits	–	●	○
DARES (2026)	Benefit recipients, 2022 cohort; logit with controls	wage employment 9 points lower in the first year than for other claimants; gap largely unexplained	–	–	●

Notes: Obj. 1: reducing segmentation; Obj. 2: defusing separations; Obj. 3: securing mobility. ● objective directly examined; ○ indirectly examined; – not examined. For Obj. 1, ○ denotes evidence on separation flows only, not on the divide between permanent and temporary jobs.

Defusing separations This objective has received the most attention, and the evidence largely runs against it. SMAs themselves are rarely litigated, but they have replaced few dismissals: only about 10–12% of dismissals for personal reasons were converted into mutual agreements (Batut and Maurin, 2019; Carry and Schoefer, 2024), and labour-court cases following such dismissals did not decline (Carry and Schoefer, 2024). SMAs appear instead to have substituted mainly for resignations (Milin, 2018; Carry and Schoefer, 2024), although the magnitude of this substitution is debated. Survey evidence also qualifies the consensual nature of the scheme: 29% of signatories report having felt compelled to leave (Bourieau, 2013).

Securing mobility The third objective is the least documented, and the existing evidence is descriptive. Access to unemployment insurance is the main reason signatories give for choosing an SMA (Bourieau, 2013). Among benefit recipients, those entering after an SMA are about 9 percentage points less likely to hold wage employment during the first year than other claimants, a gap largely unexplained by observable characteristics, although it narrows for durable employment (DARES (2026)); they are also twice as likely to start a business within two years (Unédic, 2026). These comparisons are restricted to workers who claim benefits and are drawn against claimants who lost their job through other routes. They do not identify what the SMA changes for the worker's subsequent career, nor do they inform on the quality of the next job, on wages, or on longer-run trajectories.

As Table 2 makes clear, the literature has concentrated on firms and on separation margins: how many separations the SMA created, and which ones it replaced.

1.3 Research question

This paper addresses the gap identified above by focusing on the worker's side of the 2008 wager. The 2026 retrenchment rests on an interpretation of a descriptive fact: SMA claimants remain on benefits longer than other claimants. This fact is consistent with two very different readings. Under the first, which underpins the reform, the SMA gives workers who chose to leave an insured exit, and the generosity of unemployment insurance slows their return to work. Under the second, which corresponds to the promise of 2008, the SMA allows workers to leave jobs they would otherwise have kept or quit without insurance, and longer search spells reflect a costlier but more productive transition. Distinguishing between these two readings requires comparing SMA signatories with workers who left their job through other routes, and following them beyond the end of the benefit spell.

1.4 Main methods

Identifying the effect of the SMA on career trajectories is complicated by selection. Workers who open benefit rights after an SMA differ markedly from other claimants: they are more often managers (26% against 15%) and more often hold a higher-education degree (49% against 38%) (DARES (2026); Unédic, 2026). If these workers have better prospects on the labour market to begin with, it would be unsurprising to see them reach stable jobs more often. Observed trajectories after an SMA may therefore reflect two distinct channels: the effect of the separation route itself, and the selection into the SMA of workers who already had good prospects.

To separate these two channels, I combine three approaches, all drawing on MiDAS (*Minima sociaux, droits d'assurance chômage et parcours salariés*), which links unemployment insurance records to individual employment histories.

First, I use propensity score matching to build a relevant control group of [workers who opened benefit rights after a dismissal / other separation route]. Because MiDAS records each worker's employment history before the separation, I can match not only on socio-demographic and job characteristics (age, gender, education, occupation, tenure, wage, firm size, sector, region) but also on past employment trajectories, which captures part of the differences in labour market prospects that are usually unobserved.

Second, I estimate the effect of signing an SMA on career trajectories with an event-study design on the matched sample. Under the assumption that, conditional on these characteristics and past trajectories, the separation route is as good as random, the estimates identify the causal effect of the SMA. The event-study framework also allows us to test this assumption: the absence of differences between the two groups before the separation supports the validity of the control group. Since MiDAS follows individuals for six years after the separation, I distinguish short-run effects, around the benefit spell, from longer-run effects on (employment, contract type, wages, ...).

Third, I look beyond average effects. Using sequence analysis, I classify post-separation trajectories into groups of typical career paths and examine how SMA signatories and matched controls are distributed across them. This reveals whether the SMA shifts workers towards particular

types of transitions, which an average effect typically masks.

Fourth, administrative data record what workers do after an SMA, but not why they signed it or what they would have done otherwise. To document these two dimensions, I complement the analysis with a topic model estimated on about 1,700 posts in which workers discuss their SMA on French online forums (Section 7.2).

2 Data and sample construction

I use MiDAS (*Minima sociaux, droits d'assurance chômage et parcours salariés*), an administrative database produced by the statistics department of the Ministry of Labour (DARES). MiDAS links three sources, covering the period 2017–2025:

- **Unemployment insurance records** from the national public employment service (France Travail);
- **Minimum income benefit records** from the national family allowance fund (CNAF);
- **Wage employment records** from the labour turnover files (*mouvements de main-d'œuvre*), built from employers' social declarations (*déclarations sociales nominatives*).

MiDAS includes every individual registered with France Travail for at least one day over this period, as well as every member of a household that received a minimum income benefit at least once. Its main strength for this study is that, once an individual meets either condition, their entire trajectory over the period is observed month by month across wage employment, unemployment benefit receipt and minimum income support, including the months before they entered the database. This makes it possible to observe workers both before their separation, which is needed to build a comparable control group, and for several years after it, which is needed to trace their subsequent trajectories.

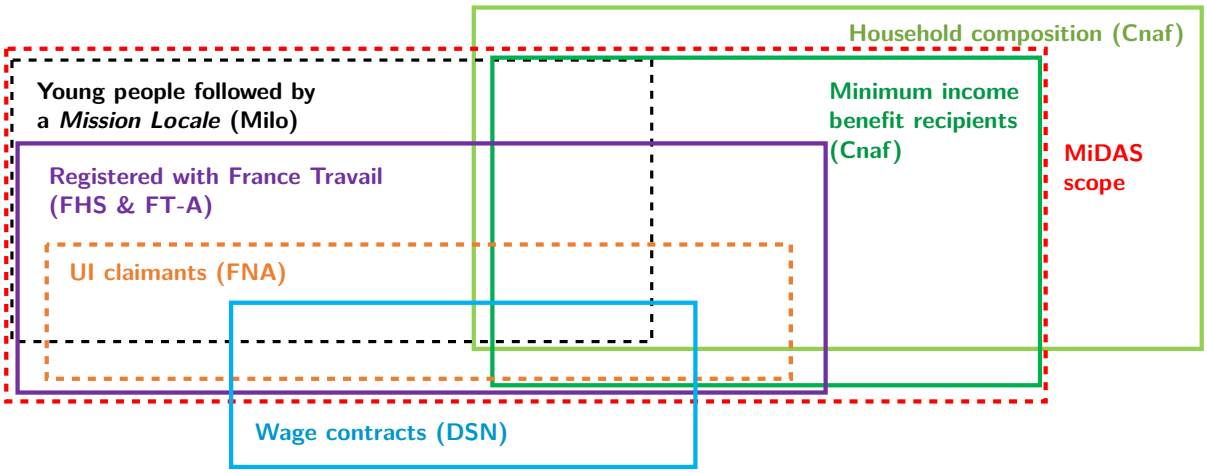


Figure 4: Scope of the MiDAS database

Notes: The MiDAS scope (red dashed line) includes individuals registered with France Travail at least one day since 2017, members of households receiving a minimum income benefit, and young people followed by a *Mission Locale*. Salaried contracts (DSN) and household composition (Cnaf) are linked for individuals within this scope. UI claimants: individuals eligible for or receiving unemployment insurance benefits. *Source:* adapted from the MiDAS documentation (DARES).

2.1 Sample construction

The unit of observation is an individual who opened an initial entitlement to unemployment insurance benefits (*allocation d'aide au retour à l'emploi*, ARE) following the end of a contract.[2]

For each month of the period under study, I identify in MiDAS all workers whose wage contract ended and gave rise to such an entitlement. For each of them, I collect four types of information:

- the **contract that generated the entitlement** (type, start and end dates, reason for termination);
- **socio-demographic characteristics** at the end of that contract;
- **subsequent employment spells**;
- **unemployment benefit receipt and minimum income support**.

Individuals are then classified according to the reason for the termination of the contract that generated their entitlement:

- **Treated:** separations by mutual agreement (SMAs).
- **Controls:** other terminations of OECs that give access to unemployment insurance (e.g. dismissals for personal or economic reasons and terminations during the trial period.)
- **Descriptive statistics only:** all other claimants, whose entitlement follows the end of a contract other than an OEC (fixed-term, temporary agency or apprenticeship contracts). They are used to situate SMA signatories within the overall claimant population but play no role in the main specification

I further build two cohorts using the same procedure.

- **The 2022 cohort** includes contracts ending between January and December 2022, followed until December 2025, which gives at least three years of follow-up. It is used for the medium-term causal estimates and allows a direct comparison with DARES (2026), who study the same cohort. It also has the advantage of lying outside the Covid-19 period.
- **The 2019 cohort** includes contracts ending between January and December 2019, followed until December 2025, which gives at least six years of follow-up. It is used for the long-term causal estimates, for benefit consumption over time and for the sequence analysis. Its trajectories cross the 2020–2021 health crisis. Because treated and control individuals are observed over the same calendar months, this shock is absorbed by the stacked design.

[2] An initial entitlement is distinguished from three other ways in which an individual may start receiving benefits, which are excluded from the sample. First, a *resumption* (*reprise de droits*): an individual who deregistered from France Travail may resume a previously opened entitlement, provided that it has not been fully used and has not reached its expiry date (*délai de déchéance*). Second, a *recharge* (*rechargement des droits*): an individual who has exhausted an entitlement may have it recharged on the basis of periods worked in the meantime. Third, an *option right* (*droit d'option*): an individual who has worked enough after deregistering to qualify for a new entitlement, while the previous one has not yet expired, may choose between the two.

2.2 Outcome variables

All outcomes are measured monthly, relative to the month in which the contract that generated the entitlement ended ($k = 0$). Table 3 defines each of them.

Table 3: Outcome variables

Outcome	Definition
Wage employment	Holds at least one hour of wage contract during the month (MiDAS definition)
Durable wage employment	Holds an OEC or a fixed-term contract of more than six months during the month
Non-salaried activity	Has reported the start of a non-salaried activity (self-employment or business creation) to France Travail
Overall employment	Wage employment or non-salaried activity
Training	Receives the training allowance (AREF, *allocation d'aide au retour à l'emploi formation*) during the month
Relative wage	Base monthly wage (*salaire de base*) of the wage contract held in month k, divided by the base monthly wage of the contract that generated the entitlement

Notes: Wage employment excludes contracts with private household employers and contracts declared through the simplified agricultural employment voucher (TESA).
Non-salaried activity: the start date is observed, but the end of the activity is only recorded while the individual receives benefits; the variable is therefore best interpreted as business creation during the entitlement period.
Training: the AREF replaces the standard benefit (ARE) for claimants who follow a training programme approved by France Travail. Only training undertaken while receiving unemployment benefits is observed; training financed by other means (e.g. the personal training account alone) is not.
The base wage is the usual gross pay, excluding all supplements, whether statutory, collectively agreed or granted by the employer (seniority, holiday and family bonuses, 13th-month pay, benefits in cash and in kind, gratuities, tips, and pay for additional or overtime hours); it generally corresponds to the first line of the payslip.

3 Who signs an SMA? From descriptive gaps to matched comparisons

This section builds on the descriptive evidence available in the literature to construct the comparison used for causal estimation. I first replicate the existing descriptive result, then examine how it changes with the choice of comparison group, and finally document the differences in characteristics that a causal comparison needs to account for.

3.1 Does the comparison group matter ?

I start by replicating the result of DARES (2026), which compares SMA claimants with all claimants of the 2022 cohort. Figure 5 confirms this finding: SMA claimants are less often in wage employment than other claimants during the first year after the end of their contract, with a gap of about 9 percentage points at $k = 12$.

All claimants, however, are not a natural benchmark. A large share of them enter the unemployment benefit rolls at the end of a fixed-term or temporary contract: at the beginning of 2026, 27% of new entrants did so after a fixed-term contract and 11% after a temporary agency assignment (Unédic, 2026b). Many of these workers alternate between short contracts and benefit receipt, so that they return to wage employment quickly, but often in short jobs. A more relevant benchmark for SMA signatories is the group of workers who also left an OEC, but through another route.

Changing the comparison group reverses the picture. Compared with workers who left an OEC through a dismissal or a termination during the trial period, SMA signatories return to wage employment more often, and reach durable employment more often, from month 3 onwards (Figure 5, right panel). The conclusion drawn from the descriptive evidence thus depends entirely on the comparison group. This raises a further question: are workers who leave an OEC through an SMA comparable to those who leave it through another route?

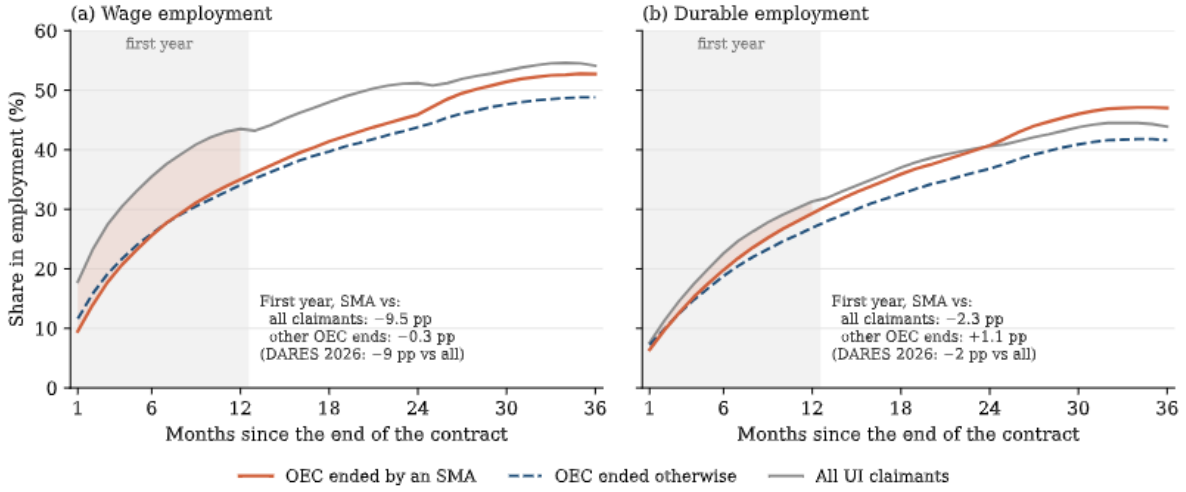


Figure 5: Share of unemployment insurance claimants in wage employment after the end of a contract (2022 cohort)

Notes: Share of individuals in wage employment in each month k after the end of the contract that generated the entitlement. *Scope:* France; individuals eligible for ARE, excluding entertainment-industry workers covered by Annexes VIII and X, whose contract ended in 2022 and was followed by an initial entitlement. *Source:* MiDAS, wave 8; Unédic calculations.

3.2 Who signs a SMA?

Table 4 answers this question by comparing the three groups at the end of the contract that generated the entitlement.

Table 4: Characteristics of claimants at the end of the contract, by separation route (2022 cohort)

	All claimants	SMA signatories	Other OEC separations
Gender			
Women	47.8	48.2	48.3
Men	52.2	51.8	51.7
Age at the end of the contract			
Under 25	25.3	7.9	12.4
25–34	28.9	34.3	27.5
35–44	20.9	27.9	24.7
45–52	12.7	15.1	16.7
53–54	2.8	3.1	3.9
55 and over	9.4	11.6	14.8
Education			
Below upper-secondary degree	47.5	38.3	52.1
Upper-secondary degree (*baccalauréat*)	20.3	18.9	19.3
Higher education	31.5	42.2	27.8
Missing	0.7	0.6	0.7
Quarter of contract end			
2022-Q1	19.3	23.4	22.5
2022-Q2	21.9	24.5	23.9
2022-Q3	32.6	24.9	26.7
2022-Q4	26.2	27.2	26.9
Daily reference wage (mean, €)	58.8	83.2	70.6
Observations	1,565,770	303,073	476,873

Notes: Column percentages unless otherwise stated. Individuals who opened an initial ARE entitlement following the end of a contract in 2022. "Other OEC separations" includes dismissals for personal or economic reasons and terminations during the trial period. *Source:* MiDAS, author's calculations.

SMA signatories do not differ from other claimants in gender, but they do in age and education. They are concentrated in the core working ages: 62% are aged 25 to 44, against 52% of other OEC leavers and 50% of all claimants. Young workers are rare among them (8% are under 25, against 12% and 25%), and so are older workers: 15% are aged 53 or over, against 19% of other OEC leavers. SMA signatories are also markedly more qualified: 42% hold a higher-education degree, against 28% of other OEC leavers and 32% of all claimants, a gap consistent with the profile documented by DARES (2026).

The timing of separations differs mainly for the claimant population as a whole, which peaks in the third quarter (33%), likely reflecting the seasonal end of fixed-term contracts. SMAs and other OEC separations are spread more evenly over the year.

The largest differences concern unemployment insurance entitlements, which reflect past employment and wages. SMA signatories have more continuous employment histories: 68% held a single contract over the two years before the separation, against 37% of all claimants. As a result, they are entitled to longer benefits: 77% can receive benefits for at least 18 months,

against 65% of dismissed workers and 48% of all claimants (DARES, 2026).[3] Their benefits are also higher, since their daily reference wage is on average €83.2, against €70.6 for other OEC leavers and €58.8 for all claimants. They also earned more before separating: their average daily reference wage is €83.2, 18% higher than that of other OEC leavers (€70.6) and 41% higher than that of all claimants (€58.8).

SMA signatories therefore enter unemployment with stronger labour market attachment and higher past earnings than workers who left an OEC through another route. Their better trajectories in Figure 5 may reflect these characteristics rather than the separation route itself. Neither naive comparison identifies the net effect of the SMA on employment outcomes.

3.3 From selection to matching

The parameter of interest is the average treatment effect on the treated at each horizon k: the difference between the outcome of SMA signatories and the outcome they would have had if their OEC had ended through another route,

$$\mathrm{ATT}_k = \mathbb{E}\big[\, Y_k(1) - Y_k(0) \mid \mathrm{SMA} = 1 \,\big], \tag{1}$$

where $Y_k(1)$ and $Y_k(0)$ denote potential outcomes k months after the separation, with and without an SMA. The second term is never observed for SMA signatories, so estimating ATT_k requires a credible counterfactual.

Ideally, this counterfactual would come from a quasi-experiment. As Table 5 shows, however, none of the usual strategies is available here.

Table 5: Quasi-experimental strategies and their applicability to the SMA

Strategy	Source of variation required	Why it is not available here
Before–after the reform	Data covering the period before and after 2008	MiDAS data only start in 2017
Differences in exposure	Regions or groups exposed to the scheme at different times or intensities	The SMA was introduced nationwide and at once
Intensity of treatment	Staggered adoption across firms or establishments, as in Batut and Maurin (2019)	This study is conducted at the level of individual workers, not firms

I therefore rely on a selection-on-observables assumption. Conditional on a rich set of pre-separation characteristics X, the separation route is assumed to be independent of the outcome a worker would have had without an SMA:

$$Y_k(0) \perp \mathrm{SMA} \mid X. \tag{2}$$

This assumption must be combined with common support: for every value of X, some workers must have left their OEC through another route, that is, $\Pr(\mathrm{SMA} = 1 \mid X) < 1$.

[3] Benefit entitlements opened in 2024. Most of these claimants are entitled to exactly 18 months (65% of SMA signatories, against 39% of all claimants), which was then the maximum duration for claimants under 53.

Under these two conditions, I use propensity score matching to build a comparison group of workers who left an OEC through another route but resemble SMA signatories in their observable characteristics.

Propensity score. The propensity score is the probability of separating through an SMA given pre-separation characteristics. It is estimated with a logistic regression:

$$e(X_i) = \Pr(\text{SMA}_i = 1 \mid X_i) = \frac{\exp(X_i'\theta)}{1 + \exp(X_i'\theta)}, \tag{3}$$

where X_i contains the variables listed in Table 6 and θ is a vector of parameters.

Matching has an advantage over regression-based adjustment: it does not extrapolate. Each SMA signatory is compared only with workers who have similar characteristics, which restricts the comparison to the region of common support. Matching does not, however, address selection on unobservable characteristics. To limit this risk, I match on detailed employment histories before the separation, and I test whether the two groups already differed before it.

Matching variables The propensity score is estimated from socio-demographic characteristics and from each individual's employment and benefit history over the 24 months preceding the end of the contract that generated the entitlement. Table 6 lists these variables.

Table 6: Variables used to estimate the propensity score

Group	Variables	Measurement
Socio-demographic characteristics	Age, gender, education, occupational skill level	At the end of the contract
Employment history	Calendar days under contract, by type of contract: OEC, fixed-term contract, temporary agency work, other contracts, non-salaried activity	Each month, over the 24 months before the end of the contract
Benefit history	Days of unemployment benefit receipt	Each month, over the 24 months before the end of the contract

Matching algorithm. Each SMA signatory is matched to the control with the closest estimated propensity score, among controls of the same gender who left their OEC in the same month. Formally, the control matched to signatory i in cohort c is

$$j(i) = \arg\min_{j \in \mathcal{C}_c(i)} \left|\hat{e}(X_i) - \hat{e}(X_j)\right| \quad \text{subject to} \quad \left|\hat{e}(X_i) - \hat{e}(X_j)\right| \leq \delta, \tag{4}$$

where $\mathcal{C}_c(i)$ is the set of controls in cohort c with the same gender as i, and $\delta = 0.05$ is the caliper. Matching is done without replacement, and signatories with no control within the caliper are dropped. Because matching is done within each monthly cohort, a treated individual and their control lose their job in the same month and face the same labour-market conditions afterwards.

3.4 Validity of the matching

I assess the quality of the matching in three steps: whether the two groups have similar means, whether they have similar dispersion, and whether they share a common support. The diagnostics below are reported for the 2022 cohort.

Balance in means Table 7 compares SMA signatories and matched controls on the matching variables. For each variable x, it reports the standardised mean difference (SMD), that is, the difference in means between the two groups divided by the standard deviation among SMA signatories:

$$\mathrm{SMD}(x) = \frac{\bar{x}_{\mathrm{SMA}} - \bar{x}_{\mathrm{C}}}{s_{\mathrm{SMA}}}, \tag{5}$$

where $\bar{x}_{\mathrm{SMA}}$ and $\bar{x}_{\mathrm{C}}$ are the means among SMA signatories and controls, and s_{SMA} is the standard deviation among SMA signatories. An absolute SMD below 0.1 is the usual threshold for negligible imbalance (Austin, 2009; Stuart, 2010).

Before matching, only 71% of the 217 matching variables had an absolute SMD below 0.1, and the largest reached 0.88. After matching, all of them fall below 0.05. The two groups are thus balanced not only on education and occupation, but also on the type and timing of employment over the 24 months before the separation. Figure 6 shows how matching reduces these differences: it plots, for each variable, the SMD before and after matching.

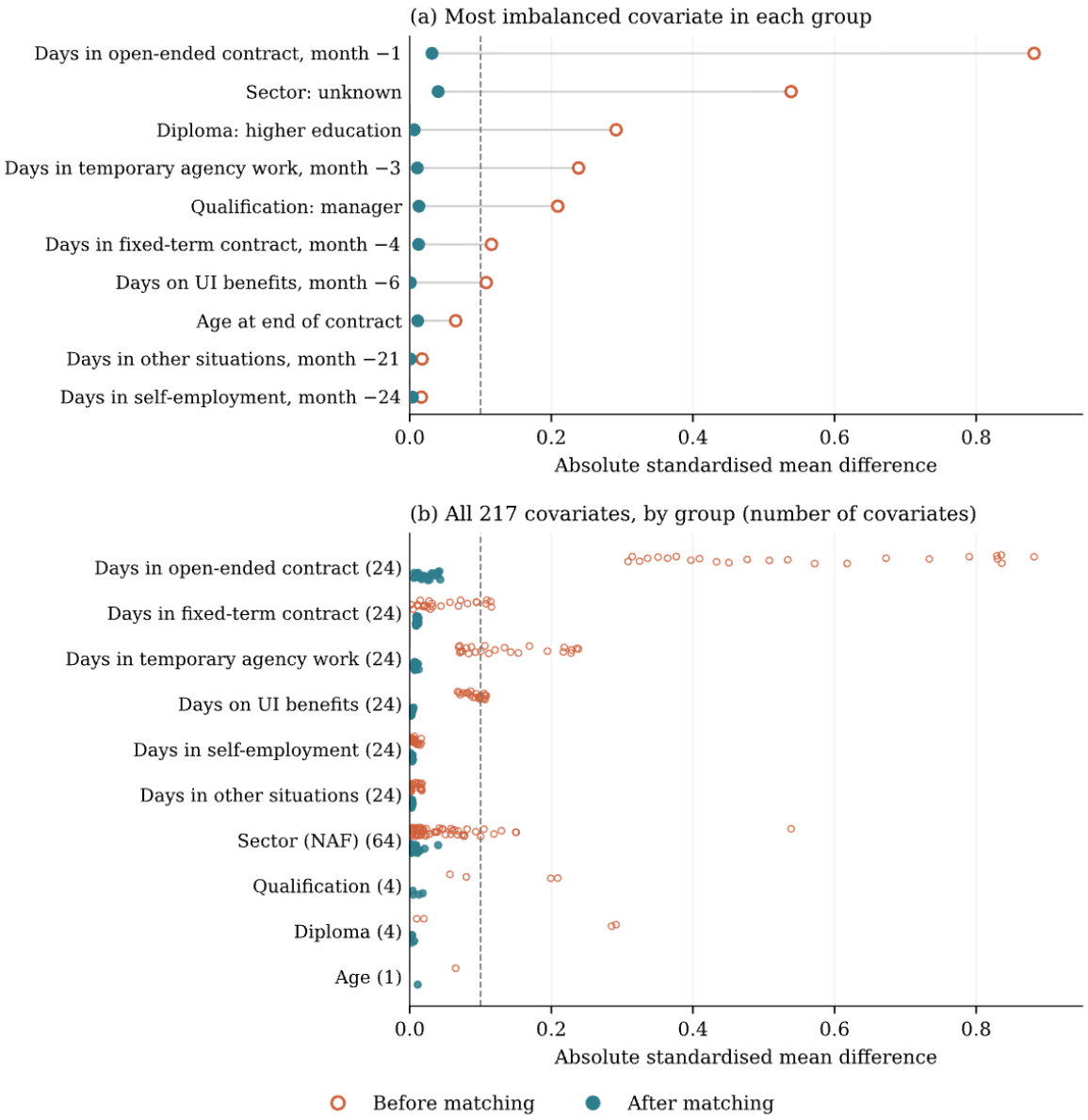


Figure 6: Covariate balance before and after matching

Notes: Absolute standardised mean differences between SMA signatories and controls, before (hollow) and after (filled) matching. (a) Most imbalanced covariate within each group of covariates. (b) All 217 covariates, by group (number of covariates in parentheses). The dashed line marks the usual 0.1 threshold. *Source:* MiDAS, 2022 cohort.

Balance in dispersion Matching also reproduces the dispersion of the matching variables among SMA signatories: the ratio of the variances in the two groups lies between 0.78 and 1.22 for all variables, close to one and well within the usual range of 0.5 to 2.[4]

[4]The only exception is a sector category (activities of households as employers) that contains almost no individuals and no matched control, for which the ratio is undefined.

Table 7: Balance before and after matching: most imbalanced variable in each group (2022 cohort)

	Before matching		After matching	
	SMD (%)	Var. ratio	SMD (%)	Var. ratio
Days under OEC, $k = -1$	88.2	0.24	3.1	0.90
Sector unknown	−53.9	0.21	−4.0	0.78
Higher education	29.1	1.21	0.6	1.00
Days in temporary agency work, $k = -3$	−23.8	0.16	−1.1	0.92
Managers	20.9	1.45	1.3	1.02
Days under fixed-term contract, $k = -4$	−11.5	0.67	1.3	1.07
Days of benefit receipt, $k = -6$	−10.8	0.34	−0.1	1.00
Age at the end of the contract	−6.5	0.83	−1.1	0.92
Days under other contracts, $k = -21$	1.7	1.38	−0.1	0.99
Days in non-salaried activity, $k = -24$	1.6	1.33	0.4	1.07
All 217 matching variables				
Largest absolute SMD	88.2		4.3	
Variables with $\|SMD\| > 10\%$	62		0	
Range of variance ratios		0.05–2.48		0.78–1.22

Notes: For each group of matching variables, the table reports the variable with the largest absolute standardised mean difference before matching. SMD: difference in means between SMA signatories and controls, in percent of the standard deviation among SMA signatories. Variance ratio: variance among SMA signatories divided by variance among controls. Days are calendar days per month; k is measured relative to the month in which the contract ended. *Source:* MiDAS, author's calculations.

Common support Finally, Figure 7 plots the density of the propensity score in each group, before and after matching. The common support assumption requires these distributions to overlap sufficiently, so that each SMA signatory can be matched with a control whose propensity score is genuinely close. The two distributions overlap substantially before matching and are nearly identical after it. Only 7% of SMA signatories fall outside the caliper and are dropped.

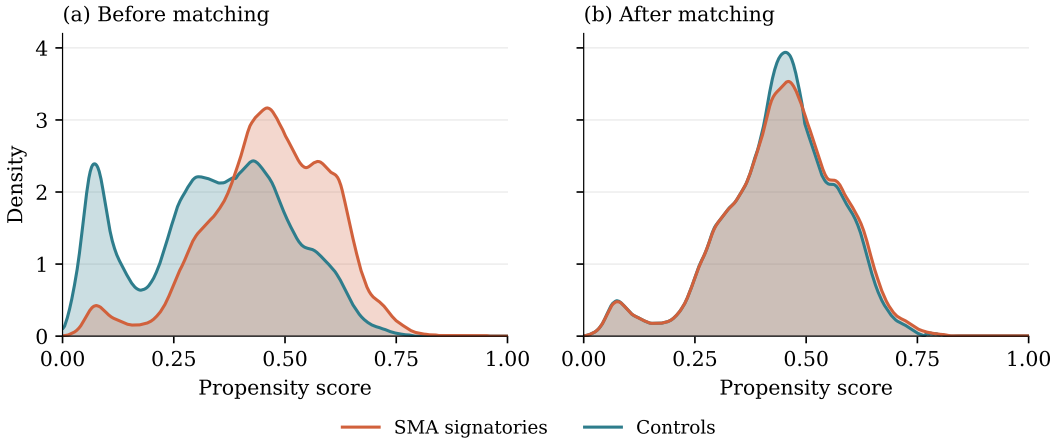


Figure 7: Density of the propensity score before and after matching (2022 cohort)

Notes: Kernel density of the estimated probability of signing an SMA, for SMA signatories and controls, before matching (upper panel) and after matching (down panel). *Source:* MiDAS, author's calculations.

4 Econometric specification

I estimate the effect of signing an SMA with an event-study design that follows individual trajectories before and after the separation. Section 4.1 describes how the sample is organised around a common event date, Section 4.2 presents the estimating equation, and Section 4.3 sets out the assumptions under which it identifies the effect of the SMA.

4.1 Event date and stacked design

Event date For each SMA signatory, the event date ($k = 0$) is the month in which the contract ended. Through the matching procedure, each control individual is assigned the same event date as the SMA signatory with whom they are matched. The trajectories of the two groups can therefore be compared relative to a common reference date.

Stacked design When treatment is staggered over time and its effects may vary with the date of entry, standard two-way fixed effects estimators can be biased, because they implicitly use units treated earlier as controls for units treated later (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Sun and Abraham, 2021). To avoid this problem, I follow a stacked event-study approach (Cengiz et al., 2019). As illustrated in Figure 8, individuals are grouped into cohorts defined by the month in which their OEC ended, each cohort containing SMA signatories and their matched controls. The regression sample stacks these cohorts, each of which constitutes a distinct quasi-experiment, so that individuals are only ever compared with others who separated in the same month.

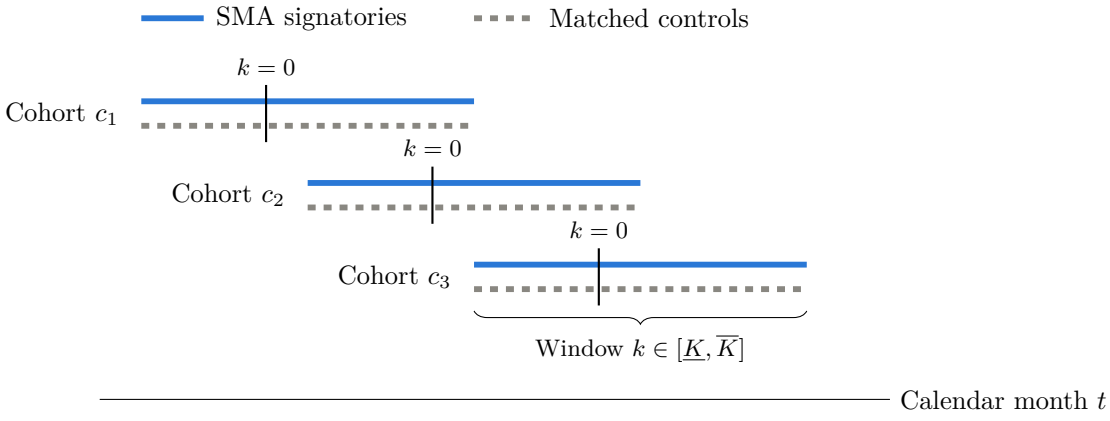


Figure 8: Stacked event-study design

Notes: Each cohort c gathers the SMA signatories whose contract ended in month c and their matched controls, who are assigned the same event date ($k = 0$). Each cohort is observed over the same event window, and the cohorts are stacked into a single regression sample. Comparisons are made only within cohorts.

4.2 Estimating equation

For individual i in cohort c observed in month t, I estimate:

$$y_{ict} = \lambda_c + \mu_i + \sum_{\substack{k=\underline{K} \\ k\neq -1}}^{\overline{K}} \gamma_k \, D_{ict}^k + \sum_{\substack{k=\underline{K} \\ k\neq -1}}^{\overline{K}} \beta_k \left(\text{SMA}_i \times D_{ict}^k\right) + \varepsilon_{ict}, \tag{6}$$

where:

- y_{ict} is the outcome of individual i in cohort c in month t (Table 3);
- SMA_i equals one if the individual separated through an SMA and zero for matched controls;
- D_{ict}^k is an indicator equal to one when the individual is observed k months from the end of the contract; the month preceding the separation ($k = -1$) is the reference period;
- β_k measure the effect of the SMA k months before or after the separation;
- γ_k capture changes common to all individuals at each distance from the event;
- λ_c are cohort fixed effects, defined by the month in which the contract ended;
- μ_i are individual fixed effects;
- ε_{ict} is an error term.

The equation is estimated separately for each cohort of the study. In both cases, the event window starts at $\underline{K} = -24$, which covers the 24 months used for matching. It ends at $\overline{K} = 36$ for the 2022 cohort, which is observed until December 2025, and at $\overline{K} = 72$ for the 2019 cohort, which allows the effects to be traced over six years.

Each component of equation (6) plays a specific role:

Cohort fixed effects (λ_c). They absorb differences in level between cohorts, such as the macroeconomic or institutional context at the time of separation. Because treated and matched control individuals within a cohort separate in the same month, they are also observed over the same calendar months: shocks occurring over the observation period, such as the business cycle, seasonality or regulatory changes, affect both groups alike. The model therefore compares treated and control individuals within the same cohort, and avoids comparisons across cohorts that could be biased by differences in composition or by aggregate shocks.

Individual fixed effects (μ_i). They control for unobserved characteristics that are constant over time, such as skills, preferences or long-standing career patterns. Identification thus comes from differences in changes within individuals and within cohorts. They are needed because the same individual may serve as a control in several cohorts, since matching without replacement is performed within each cohort; this is not the case for treated individuals.

Inference. Standard errors are clustered at the individual level to account for serial correlation in the panel.

4.3 Identifying assumptions

The coefficients β_k identify ATT_k under two assumptions.

Conditional independence. Within the same monthly cohort and at equal propensity score, the separation route is unrelated to potential outcomes. The balance obtained on the 217 matching variables (Section 3.4) supports this assumption for observable characteristics, but it cannot rule out differences in unobservable characteristics that matter for labour market outcomes, such as ability or motivation. The individual fixed effects μ_i partly address this concern: they absorb any unobserved difference that is constant over time, so that the assumption only needs to hold for changes in outcomes rather than for their levels.

Parallel trends. In the absence of an SMA, signatories and matched controls would have followed the same trajectory relative to the reference month:

$$\mathbb{E}\big[\, Y_k(0) - Y_{-1}(0) \mid \text{SMA} = 1, c, \hat{e}\,\big] = \mathbb{E}\big[\, Y_k(0) - Y_{-1}(0) \mid \text{SMA} = 0, c, \hat{e}\,\big]. \tag{7}$$

The pre-separation coefficients β_k, for $k < 0$, provide a test of this assumption: they should be close to zero.

5 Results: Average Treatment Effect of SMAs on Employment

This section presents the estimates of β_k from equation (6). Section 5.1 reports the effect of the SMA on wage employment for the 2022 cohort, observed for three years after the separation. Section 5.2 explains this effect by distinguishing durable employment and non-salaried activity. Section 5.3 uses the 2019 cohort to examine whether these effects persist over six years.

5.1 What is the effect of SMAs on wage employment ?

The SMA delays the return to wage employment, but the delay is small and temporary: SMA signatories are at most 2.4 percentage points less likely than matched controls to be in wage employment, and the gap disappears about two and a half years after the separation. Figure 9 plots the estimates month by month for the 2022 cohort.

A small initial gap. The gap opens as soon as the contract ends and is largest in the first months, at -2.4 percentage points at $k = 2$. This effect is much smaller than the raw gap between SMA claimants and all claimants (Section 3.1): most of the difference observed in descriptive comparisons reflects differences in characteristics rather than the separation route itself.

A gap that closes within two and a half years. The gap then narrows steadily, to -1.4 points after one year and -1.1 points after two years, before closing quickly. It is no longer

statistically significant from $k = 29$ onwards and is zero three years after the separation (-0.02 points at $k = 36$). The SMA therefore affects the timing of the return to wage employment, not its level in the medium run.

A convergence that coincides with benefit exhaustion. The gap narrows sharply right after $k = 24$, which corresponds to the maximum benefit duration for claimants under 53 in the 2022 cohort. SMA signatories are more often entitled to this maximum duration than other OEC leavers (Table 4). The pattern is therefore consistent with a return to work at benefit exhaustion, although the timing alone does not establish this mechanism.

Pre-separation coefficients. Before the separation, the coefficients are close to zero. Those for the six months preceding the separation are statistically significant but very small, between -0.2 and -0.3 percentage points, which reflects the precision allowed by the sample size rather than a meaningful pre-tren

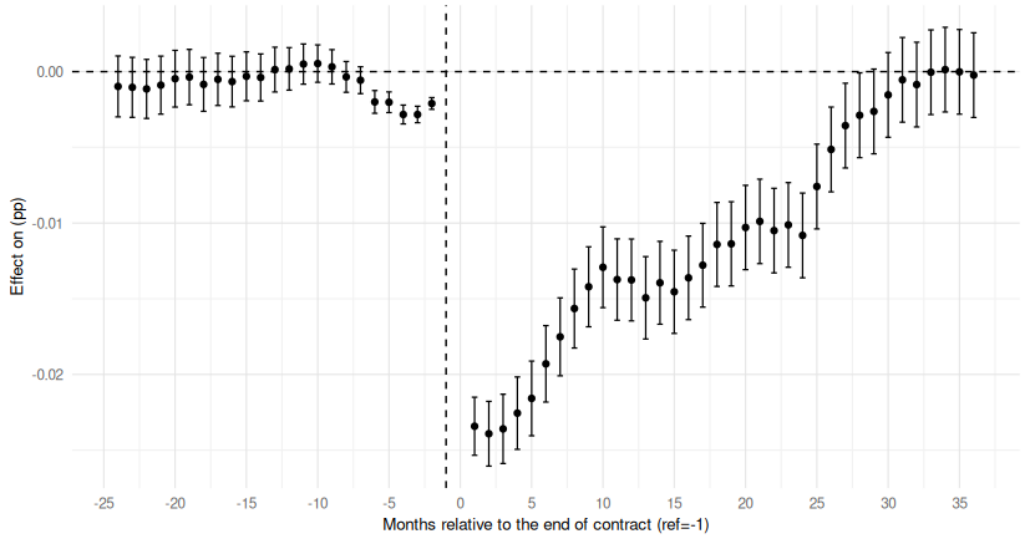


Figure 9: Effect of the SMA on wage employment (2022 cohort)

Notes: Estimates of β_k from equation (6), in percentage points, with 95% confidence intervals. The outcome equals one if the individual holds at least one salaried contract in month k. The reference period is $k = -1$ (hollow marker). The dotted line marks 24 months after the separation. Standard errors clustered at the individual level. *Source:* MiDAS, author's calculations.

5.2 What explains the gap? Self-employment, training and job selectivity

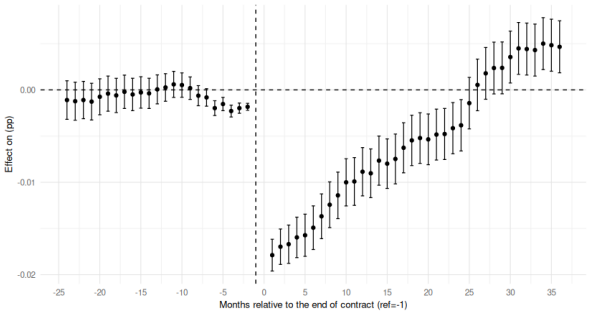
The lower rate of wage employment among SMA signatories does not necessarily mean that they spend more time out of work. Two mechanisms can account for it: some signatories leave salaried employment for self-employment, and those who return to salaried employment are more selective about the jobs they accept. Figure 10 documents both.

Transitions to self-employment absorb most of the gap. Panel (b) shows that SMA signatories are much more likely to start a non-salaried activity than matched controls. The

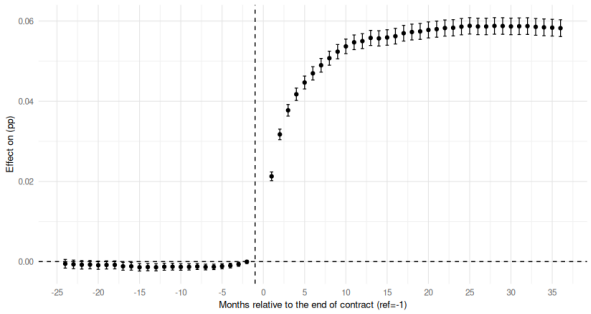
effect reaches 2.1 percentage points one month after the separation, 5.5 points after one year, and then stabilises at around 5.8 points. This effect is of the same order as the gap in wage employment from the first month (-2.3 points at $k = 1$), and larger thereafter (-1.4 points at $k = 12$). Because the end of a non-salaried activity is only observed during benefit receipt, this estimate is an upper bound, and some individuals may combine both activities. The comparison nonetheless suggests that a large part of the gap in wage employment reflects a shift towards self-employment rather than a longer period without work.

Training accounts for part of the gap in the first year. I show that SMA signatories enter training earlier and more often than matched controls. The effect peaks at 0.9 percentage points three months after the separation, when it represents almost 40% of the gap in wage employment (-2.4 points at $k = 3$). It then declines and becomes slightly negative from the twelfth month onwards, at around -0.1 to -0.2 points. Part of the time SMA signatories spend out of salaried work in the first year is therefore devoted to training: they bring forward an investment in skills that controls make, if at all, later.

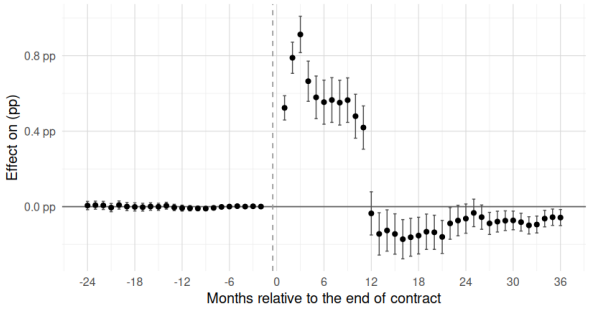
The remaining gap reflects fewer short jobs. Panel (a) decomposes wage employment by type of job. The gap in durable employment, defined as an OEC or a fixed-term contract of more than six months, is smaller than the gap in wage employment (-1.8 points at $k = 1$) and closes faster: -0.9 points after one year and -0.4 points after two years. It then reverses: from $k = 30$ onwards, SMA signatories are significantly more likely than matched controls to hold a durable job, by about 0.5 points at the end of the third year. By difference, SMA signatories hold fewer short jobs throughout the period, by about 0.5 points. Over time, the gap in wage employment thus shifts from durable to short jobs: SMA signatories return more slowly, but are less likely to accept short contracts and more likely to end up in durable jobs.



*(a) Durable employment



*(b) Non-salaried activity



*(c) Training

Figure 10: Effect of the SMA by type of activity (2022 cohort)

Notes: Estimates of β_k from equation (6), in percentage points, with 95% confidence intervals. Panel (a): the outcome equals one if the individual holds an OEC or a fixed-term contract of more than six months in month k. Panel (b): the outcome equals one if the individual has reported the start of a non-salaried activity to France Travail; since the end of this activity is only observed during benefit receipt, the estimate is better understood as business creation during the benefit receipt period. Panel (c): the outcome equals one if the individual receives the training allowance (*allocation d'aide au retour à l'emploi formation*, AREF) in month k. The reference period is $k = -1$. *Source:* MiDAS, author's calculations.

Interpretation. The three mechanisms point to a more selective transition rather than a weaker attachment to the labour market. SMA signatories spend part of the time out of wage employment starting a business or training, and those who return to wage employment take up fewer short jobs and more durable ones. This pattern is consistent with the role of unemployment insurance in financing the transition: by preserving benefit entitlements, the SMA gives signatories the time to start a business, to train, or to wait for a better match, rather than taking up the first short job available.

5.3 Do the effects persist in the long run?

The 2022 cohort can only be followed for three years. To assess whether the convergence observed after 24 months is durable, we turn to the 2019 cohort, which can be followed for six years ($\overline{K} = 72$). Estimates are quarterly and the reference period is $k = -3$.

5.3.1 Wage employment over six years

The first three years confirm the medium-run pattern. Over the first three years, the 2019 cohort reproduces the profile observed for 2022: a negative gap in wage employment after the end of the contract, which narrows sharply once the 24-month maximum benefit duration is reached (Figure 11).

A small gap reappears in the longer run. Beyond three years, the convergence is not complete. From $k = 39$ onwards, SMA signatories are about 0.5 percentage point less often in wage employment than their matched controls, and this gap persists until the end of the observation window. It is small, but it is stable.

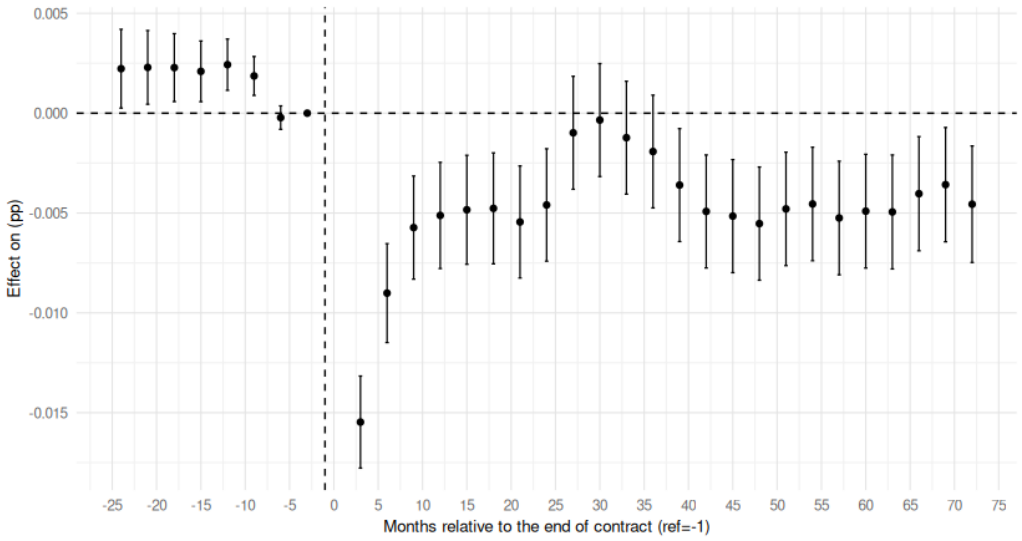


Figure 11: Effect of the SMA on wage employment over six years (2019 cohort)

Notes: Stacked event-study estimates of β_k (quarterly), reference period $k = -3$, with 95% confidence intervals. Standard errors are clustered at the individual level. *Source:* MiDAS, 2019 cohort.

5.3.2 Including self-employment

The persistent wage employment gap does not mean that SMA signatories are more often out of work. Over the same horizon, they are 5.1 percentage points more likely to report a non-salaried activity than their matched controls, an effect ten times larger than the wage employment gap. Unless most of these activities had ceased, SMA signatories are therefore likely more often in work overall: part of them leave wage employment for self-employment, a transition the SMA, which preserves UI entitlements, makes easier to finance.

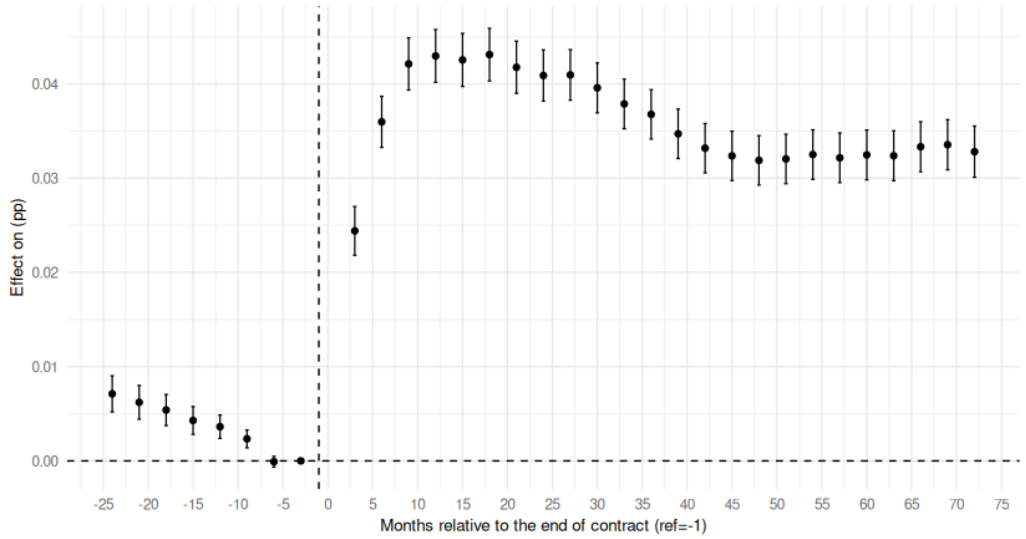


Figure 12: Effect of the SMA on total employment over six years (2019 cohort)

Notes: Stacked event-study estimates of β_k (quarterly), reference period $k = -3$, with 95% confidence intervals. Total employment: wage employment or non-salaried activity. Standard errors are clustered at the individual level. *Source:* MiDAS, 2019 cohort.

5.3.3 Wages at re-employment

Returning to work says little about the quality of the jobs found. For each individual in wage employment in month k, I compute the ratio of the base wage of the contract held that month to the base wage of the contract that ended (Table 3). A ratio of one means the same base wage as before. The base wage is only part of total pay: it excludes bonuses, overtime and other supplements, and accounted for about 75% of gross pay in private firms with ten or more employees in 2020 (DARES, n.d.). My measure therefore captures the core of the job's pay, not total earnings.

Figure 13 reports the median of this ratio, that is, the value below which half of re-employed individuals fall.[5] The two groups follow almost the same path. One year after the end of the contract, the median signatory earns a base wage about 4% higher than before, as do matched controls. The median ratio then rises at the same pace in both groups, to about 1.18 after six years.

SMA signatories who return to wage employment therefore do not take lower-paid jobs than comparable workers. This is consistent with the idea that many SMAs correspond to chosen job changes rather than forced exits: signatories do not suffer the wage losses usually found after involuntary job loss, and those who return to work do so on terms similar to other workers who left an OEC.

[5] I do not estimate the event-study model on this outcome. The ratio is only defined for individuals in wage employment in month k, and the separation route affects who is employed and when. Comparing wages among the employed would therefore compare groups that are no longer matched: if, for example, signatories with the best prospects return to work later, their wages enter the comparison at different horizons than those of controls. The results in Figure 13 should thus be read as descriptive statistics on those in employment, not as the effect of the SMA on wages.

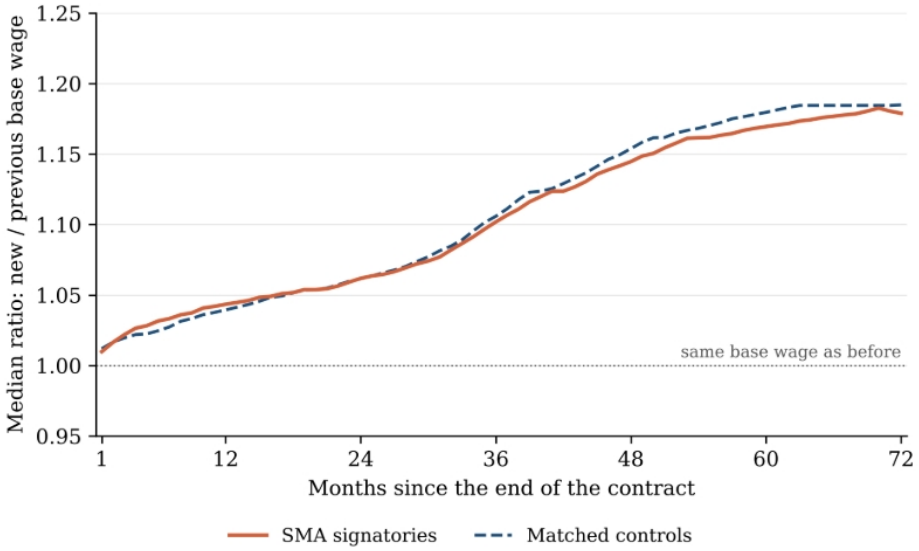


Figure 13: Median base wage of the new contract relative to the previous one

Notes: Median, among individuals in wage employment in month k, of the ratio of the monthly base wage of the contract held in month k to the monthly base wage of the contract that ended. *Source:* MiDAS, 2019 cohort, matched sample.

5.3.4 Cumulative benefit consumption

The 2026 reform rests on a cost argument: by granting UI to workers who leave their job by mutual agreement, the SMA would lead to longer, and more costly, benefit spells. The employment results speak only indirectly to this argument, since a lower employment rate does not necessarily mean more benefit days. I therefore measure UI consumption directly, as the cumulative number of benefit days paid from the end of the contract up to horizon h, for SMA signatories and their matched controls in the 2019 cohort.

SMA signatories consume slightly more benefits in the first two years. During the first two years, SMA signatories accumulate slightly more benefit days than their matched controls (Figure 14 and Table 8). The gap peaks at 24 months, with 5.8 additional days, i.e. 1.4% of the controls' cumulative consumption. This peak coincides with the maximum benefit duration.

The gap closes after three years and then reverses. After 24 months, the gap narrows quickly. Cumulative consumption is equal at about 36 months, and then becomes lower for SMA signatories. After six years, they have received 10.7 fewer benefit days than their matched controls (609.6 against 620.3, i.e. −1.7%). Over the whole period, SMA signatories therefore do not cost the UI system more than comparable workers, and they end up costing slightly less.

Interpretation. This reversal is consistent with the exit of some SMA signatories from the UI system towards self-employment (Section 5.3.2). Average cumulative outcomes, however,

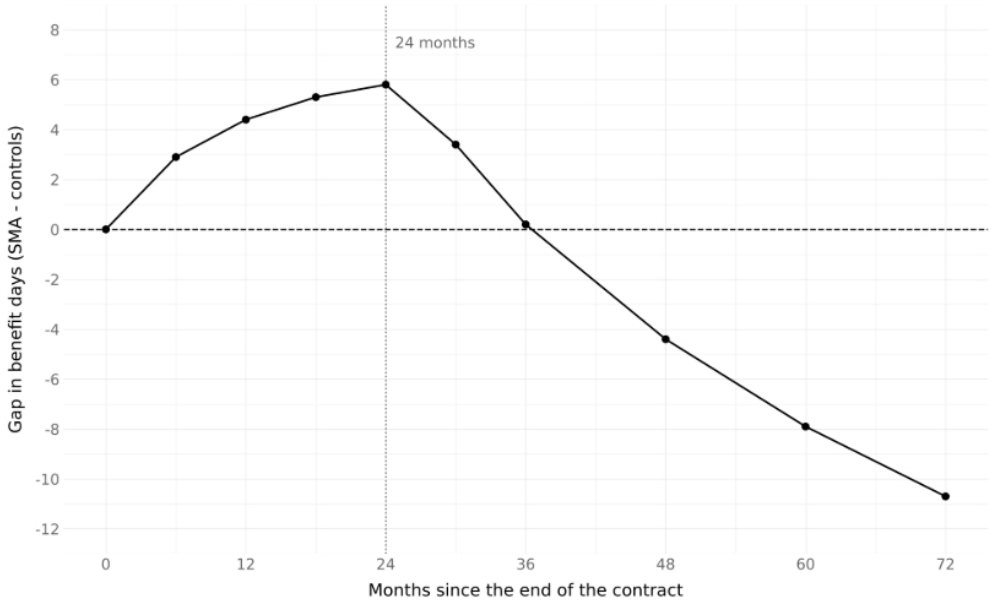


Figure 14: Gap in cumulative benefit days between SMA signatories and matched controls (2019 cohort)

Notes: Difference in mean cumulative UI benefit days between SMA signatories and their matched controls, from the end of the contract to each horizon (in months). The dotted line marks the 24-month maximum benefit duration. $N = 254{,}418$ individuals per group. *Source:* MiDAS, 2019 cohort.

cannot tell us how these differences build up over time. A similar number of benefit days can result from very different trajectories: one long benefit spell followed by stable employment, or repeated movements between short jobs and unemployment. The gap between the mean and the median, which stays slightly positive (+2 days at 72 months), also suggests that the reversal is driven by a subset of individuals rather than by a shift of the whole distribution.

Table 8: Cumulative benefit days: mean and median gaps between SMA signatories and matched controls (2019 cohort)

Horizon (months)	12	24	36	60	72
Mean gap	+4.4	+5.8	+0.2	−7.9	−10.7
Median gap	+11	+16	+13	+5	+2

Notes: Gap = SMA signatories − matched controls, in UI benefit days cumulated from the end of the contract to each horizon. $N = 254{,}418$ individuals per group. *Source:* MiDAS, 2019 cohort.

To identify which trajectories lie behind these averages, and why matched controls end up consuming more benefits than SMA signatories, we turn to a sequence analysis of individual labour market paths (Section 6). This approach describes the full succession of states each individual goes through after the end of the contract, rather than the share of individuals in a given state at each date.

6 Beyond the average effect: a sequence analysis

The event study gives an average effect at each horizon, but the same average can hide very different individual paths. A small deficit in wage employment may reflect a short delay for everyone, or a large change of trajectory for a minority. Similarly, the lower long-run benefit consumption of SMA signatories (Section 5.3.4) may come from a few distinct paths rather than from a general shift. This section therefore looks at heterogeneity. It asks two questions: which paths lie behind the small wage employment gap, and why matched controls end up consuming more benefits than SMA signatories. I use the 2019 cohort, which offers the longest follow-up.

6.1 Methodology : From individual trajectories to typical paths

6.1.1 A stratified sample of matched pairs

Sequence analysis requires computing a distance between every pair of trajectories, which is not feasible on the full matched sample. I therefore draw a sample of 20,000 matched pairs, stratified on the characteristics of the SMA signatory: sex, qualification, diploma and duration category, crossed with the quartiles of age at the end of the contract and of the daily reference wage. Pairs are allocated to strata in proportion to their size in the matched population (largest remainder method), then drawn at random within each stratum. Because pairs are drawn as a whole, each SMA signatory keeps its matched control, and the balance obtained by matching is preserved in the sample. The distribution of the stratification variables in the sample matches that of the full matched population.

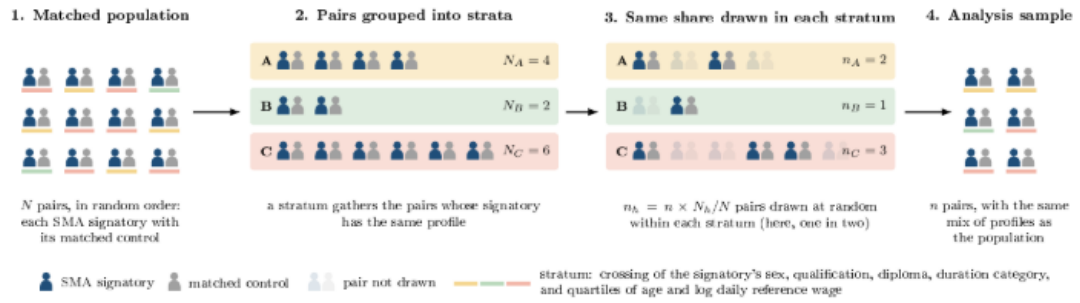


Figure 15: Stratified proportional sampling of matched pairs

Notes: Stylised illustration with three strata and a sampling rate of one in two. In the analysis, N is the number of matched pairs in the population, N_h the number of pairs in stratum h and $n = 20{,}000$ the target sample size; allocations n_h are rounded with the largest remainder method. Pairs are drawn as a whole, so that each signatory keeps its matched control. *Source:* MiDAS, 2019 cohort.

6.1.2 Trajectories as sequences of monthly states

Each trajectory is a sequence of 67 monthly states, from the first month after the end of the contract ($k = 1$) to the 67th month.[6] Each month, the individual is assigned to one of nine mutually exclusive states (Table 9).

[6]The event study follows the 2019 cohort over 72 months. The sequence analysis stops at 67 months because the CNAF data linked in MiDAS, which identify minimum income recipients, end in July 2025. Since every

Table 9: Monthly states used in the sequence analysis

State	Definition
Wage employment	Wage contract, no benefit
Wage employment with benefits	Wage contract combined with UI (reduced activity) or with a minimum income benefit
Non-salaried and wage	Non-salaried activity combined with a wage contract
Non-salaried, indemnified	Non-salaried activity combined with UI
Non-salaried only	Non-salaried activity, without UI or wage contract
Indemnified unemployment	UI benefits without any activity
Minimum income	RSA or AAH, without activity
Retirement	Exit to retirement
Not observed	None of the above recorded in the data

6.1.3 Measuring how far two trajectories are

I measure the dissimilarity between two trajectories with optimal matching (Abbott and Tsay, 2000; Lesnard, 2010): the distance between two sequences is the minimum cost of the operations needed to transform one into the other. Substituting one state for another costs 2, and inserting or deleting a state, which shifts the sequence in time, costs 1.5. The distance is thus sensitive both to the order of states and to the timing of transitions.

6.1.4 One typology for both groups, in four classes

I use a single typology for both groups, rather than constructing separate ones. This makes it possible to compare the trajectories of SMA signatories and their controls directly. The analysis then looks at how the two groups are distributed across the different paths: which paths account for the small gap in wage employment, and in which paths are controls more likely to be found, potentially explaining their higher benefit consumption in the long run? Because the typology is constructed without distinguishing between the two groups, differences in their distribution across paths can then be used to examine how the trajectories of SMA signatories differ from those of their controls.

The trajectories are then classified together with an agglomerative hierarchical clustering based on Ward's criterion, applied to the matrix of optimal matching distances between all pairs of trajectories.

monthly state must be observable for all sources, including minimum income, the trajectories are truncated at the last horizon covered by the CNAF data.

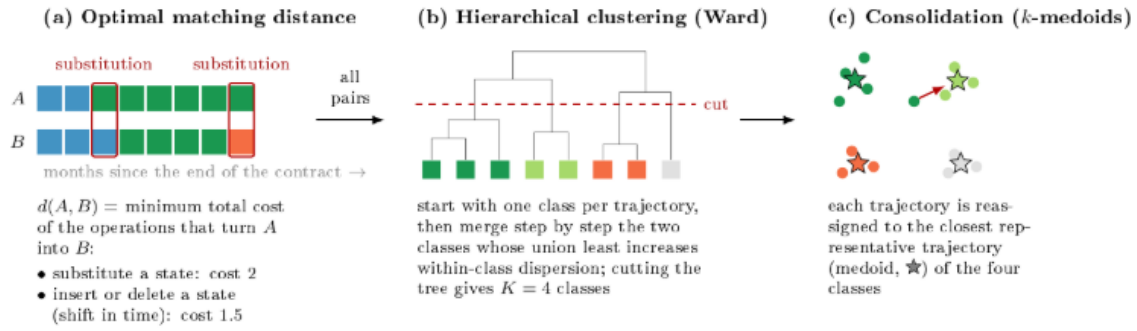


Figure 16: From individual trajectories to a typology

Notes: Stylised illustration. (a) Optimal matching distance between two monthly sequences. (b) Ward's agglomerative hierarchical clustering on the matrix of pairwise distances, cut at four classes. (c) The partition is consolidated by a k-medoids algorithm initialised with the hierarchical solution.

I retain four classes. The inertia jumps of the clustering decrease steeply up to four classes and then level off (Figure 17, panel a). The standardised quality indices (Studer, 2013) all improve from three to four classes (panel b). They keep improving beyond and reach their best values between seven and nine classes, but at the cost of many small classes, harder to interpret and too small for a reliable comparison between the two groups. Four classes are therefore a compromise between statistical quality and interpretability. The partition is finally consolidated with a k-medoids algorithm around the centres of the hierarchical solution.

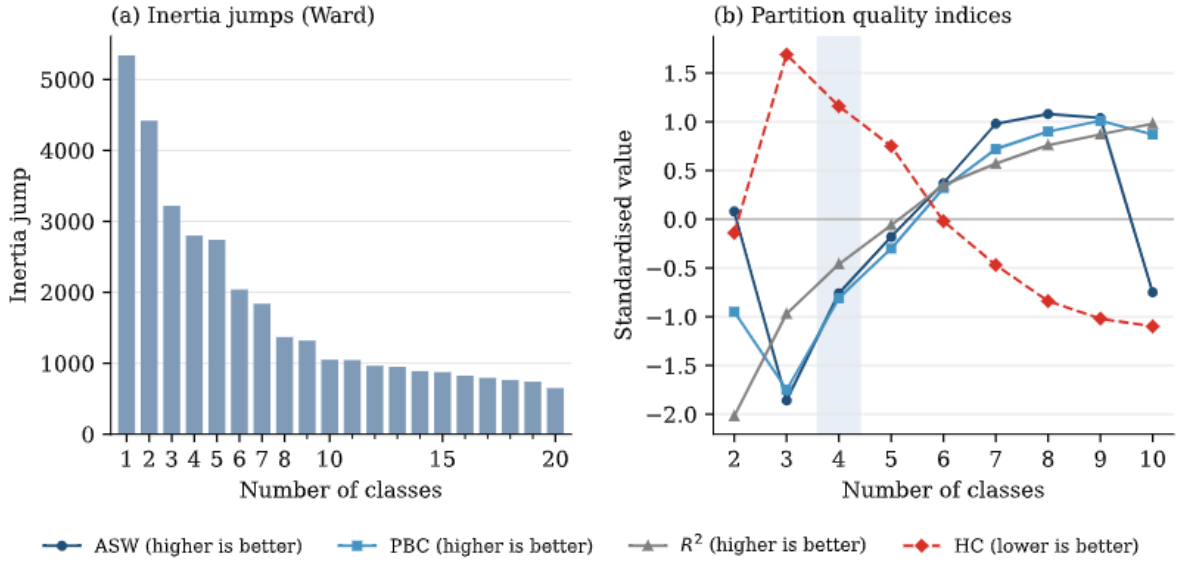


Figure 17: Choice of the number of classes

Notes: (a) Loss of between-class inertia at each step of Ward's hierarchical clustering. (b) Standardised partition quality indices: average silhouette width (ASW), point biserial correlation (PBC) and pseudo-R^2 (higher is better), Hubert's C (HC, lower is better). The shaded area marks the retained four-class solution. *Source:* MiDAS, 2019 cohort, stratified sample of matched pairs.

6.2 Results

6.2.1 Four typical paths

The clustering identifies four classes, described by their chronograms (distribution of states by month) and their most representative sequences.

1. **Fast return to wage employment.** A short period of unemployment with benefit receipt, followed by stable wage employment. This is the largest class, accounting for around 40% of individuals.
2. **Wage employment with benefits.** These trajectories combine wage employment with periods of support, either through UI (reduced activity) or a minimum income benefit. They also alternate between employment and unemployment with benefit receipt. Employment is recurrent, but often part-time, low-paid or interrupted.
3. **Self-employment.** These trajectories involve self-employment, initially alongside UI and later without it. Some individuals also hold a wage job.
4. **Exit from the observed field.** These trajectories begin with unemployment with benefit receipt and end in retirement, minimum income support, or no further record in the data. This class includes older individuals and those who leave the labour market or fall outside the scope of MiDAS.

6.2.2 Same main path, different minority paths

Most SMA signatories follow the same path as their controls: they are almost equally represented in the fast-return class (39.2% against 40.0%, Table 10). This is consistent with the event study. The wage employment gap is small and temporary (Section 5.1), and over six years it does not exceed 0.5 percentage point (Section 5.3.1). The differences are therefore not spread across the whole population, but concentrated in two minority paths.

Table 10: Distribution of SMA signatories and matched controls across typical paths (2019 cohort, %)

Typical path	SMA signatories	Controls	Difference (pp)
Fast return to wage employment	39.2	40.0	−0.8
Wage employment with benefits	23.0	26.5	−3.5
Self-employment	21.0	15.0	+6.0
Exit from the observed field	16.0	18.0	−2.0

Notes: Stratified sample of 20,000 matched pairs. Monthly states from $k = 1$ to $k = 67$. Classes obtained by optimal matching and hierarchical clustering on the pooled trajectories.
Source: MiDAS, 2019 cohort.

More self-employment. 21% of SMA signatories follow the self-employment path, against 15% of their controls, a difference of 6 points. This is close to the event-study estimate of about +5 points in non-salaried activity at six years (Section 5.3.2), obtained with an entirely

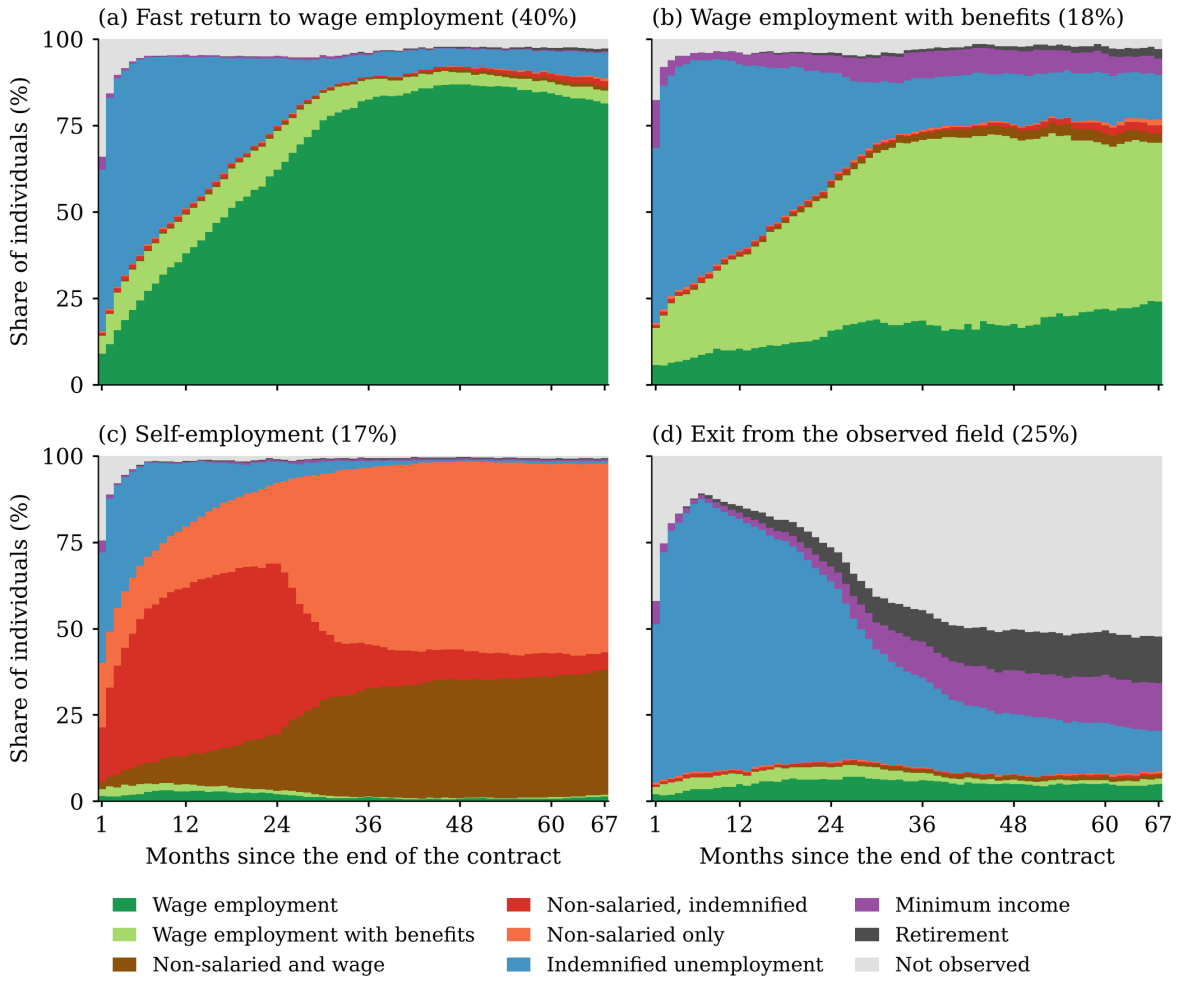


Figure 18: Chronograms of the four typical paths (2019 cohort)

Notes: Monthly distribution of states within each class, from the first to the 67th month after the end of the contract. SMA signatories and matched controls are pooled. The share of each class in the sample is given in parentheses. Classes are obtained by optimal matching and hierarchical clustering, consolidated by k-medoids.
Source: MiDAS, 2019 cohort, stratified sample of matched pairs.

different method. The self-employment path therefore accounts for most of the residual wage employment gap: SMA signatories who are missing from wage employment are, to a large extent, in self-employment.

Fewer unstable wage paths. 23% of SMA signatories follow the path of wage employment with benefits, against 26.5% of controls, a difference of 3.5 points. This path combines wage employment, wage employment that requires support, and alternation between employment and unemployment spells.

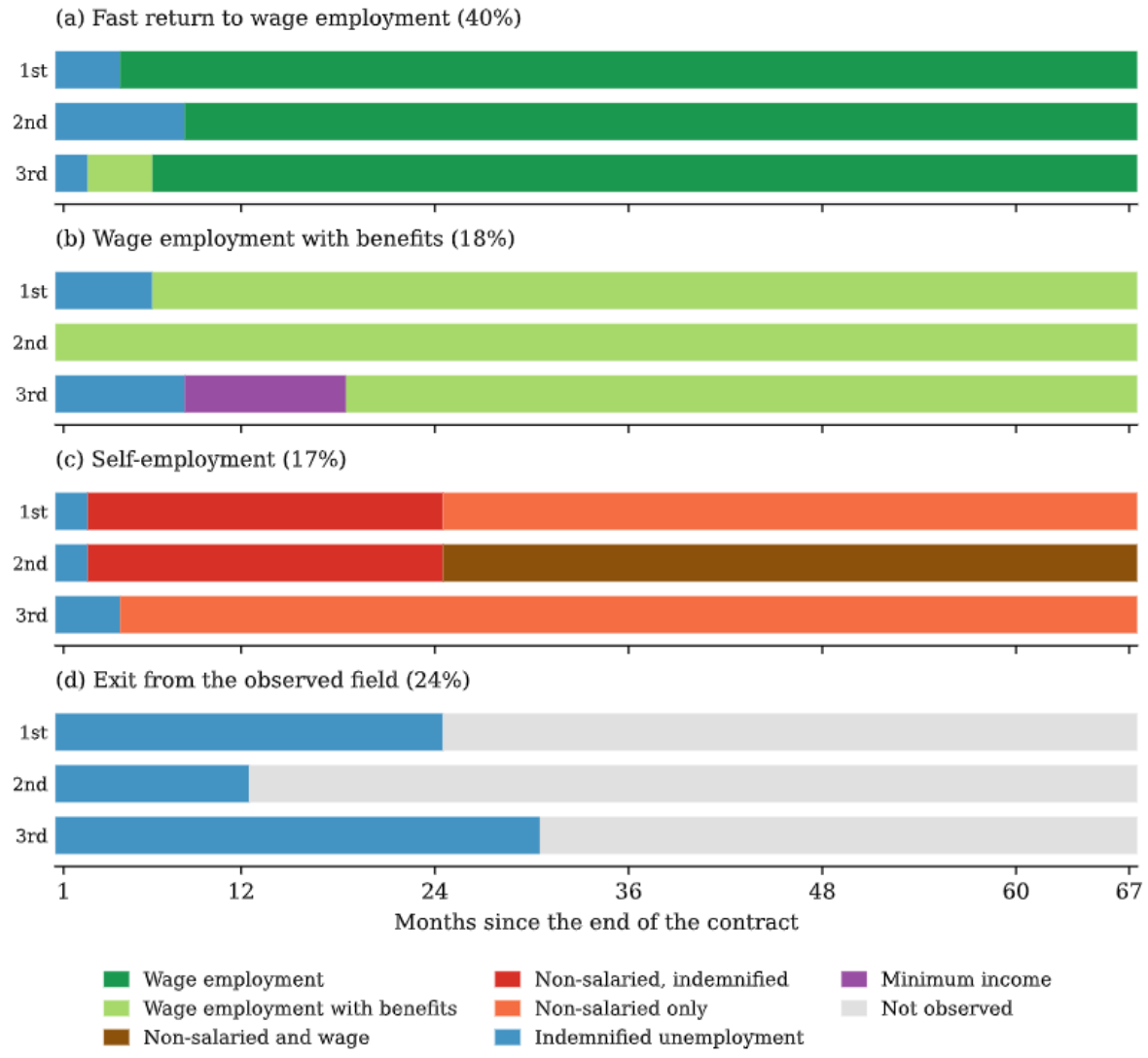


Figure 19: Most representative sequences of each typical path (2019 cohort)

Notes: For each class, the three most representative sequences according to the density criterion (sequences with the largest number of close neighbours), from the first to the 67th month after the end of the contract. The share of each class in the sample is given in parentheses. SMA signatories and matched controls are pooled. *Source:* MiDAS, 2019 cohort, stratified sample of 20,000 matched pairs.

6.2.3 Is self-employment a lasting exit?

In MiDAS, non-salaried activity is only observed while individuals are registered with France Travail, through what they declare during their benefit spell.[7]. Once they leave the register, the activity is no longer observed. I therefore treat entry into self-employment as an absorbing state: individuals remain classified as self-employed until the end of the window, whether or not the activity continues. The data thus do not tell whether the business survives, and remaining in the self-employment class can have two readings:

- **a viable activity**: individuals stay in self-employment because their business works;
- **a lack of alternatives**: individuals stay out of wage employment because they could not find a wage job.

To distinguish between these two readings, I compare the wage employment that individuals in each class could expect given their profile with the wage employment they actually have.

Predicting wage employment from the initial profile. For the self-employment class, a random forest is trained on the individuals of the three other classes and then used to predict the wage employment of self-employed individuals from their profile (Figure 20). The same procedure is then repeated for each of the other classes: each class is predicted by a model trained on the three remaining classes, and therefore never sees its own members. The predicted rate is thus the wage employment observed among individuals with a similar profile who followed another type of path.[8] Comparing predicted and observed rates then shows, for each class, whether its members did better or worse in wage employment than their profile would suggest.

[7] This limitation should be lifted in the coming years: future versions of MiDAS are expected to include data from the Acoss (*Agence centrale des organismes de sécurité sociale*), which would allow non-salaried activity to be observed independently of registration with France Travail.

[8] Training the model on the class it predicts would partly reproduce the class's own outcome: for the self-employment class, it would teach the model that these profiles do not work as employees, and would mechanically lower their predicted rate.

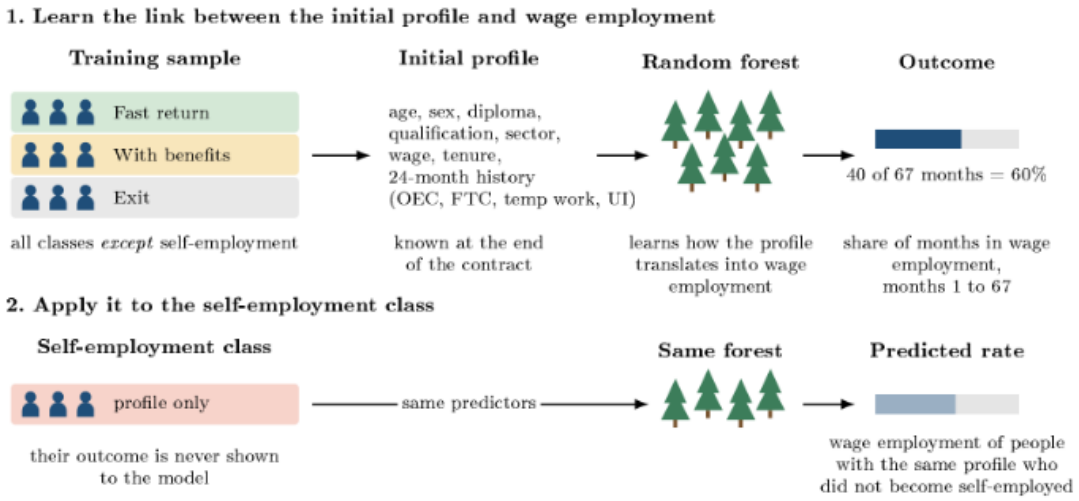


Figure 20: Predicting expected wage employment from the initial profile

Notes: Stylised illustration of the reference model. The forest is trained on all individuals outside the self-employment class (out-of-bag predictions for them) and then applied to the profile of the self-employment class. *Source:* MiDAS, 2019 cohort, stratified sample of matched pairs.

Same profile as those who return quickly to wage employment. Given their profile, individuals in the self-employment class had the same wage employment prospects as those who returned quickly to wage employment, and better prospects than the two other classes: their predicted rate is 67% at 36 months, against 61% for the class with benefits and 51% for the exit class (Figure 21, panel a). Their move into self-employment cannot therefore be explained by weaker prospects in wage employment. Yet their observed wage employment remains far below this prediction, at 38% after 60 months, 27 points less than their profile would suggest (panel b). Since they had good outside options, remaining out of wage employment is more likely to reflect a choice, and a viable activity, than a failure to find a job.

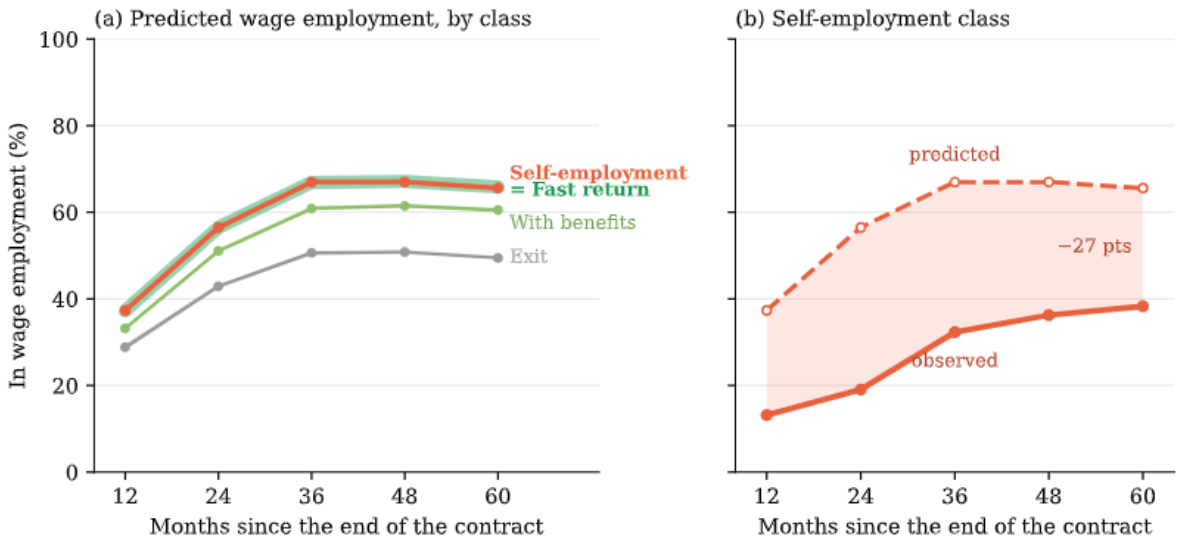


Figure 21: Predicted and observed wage employment of the self-employment class

Notes: (a) Share of individuals predicted to be in wage employment at each horizon, given their profile at the end of the contract, by typical path. (b) Predicted and observed wage employment of the self-employment class.
Source: MiDAS, 2019 cohort, stratified sample of matched pairs.

6.2.4 What this adds to the average effects

Which paths lie behind the small wage employment gap? Not a generalised delay: most SMA signatories follow the same fast-return path as their controls. The gap comes mainly from the larger share who move into self-employment. Since their profile gave them the same wage employment prospects as those who return quickly, this move looks like a choice rather than a failure to find a job.

Why do controls consume more benefits in the long run? Controls more often follow unstable wage paths combined with benefits, which involve repeated benefit receipt (Section 5.3.4). SMA signatories return less often to unemployment insurance, partly because more of them leave it for, what is believed to be more stable self-employment.

7 Discussion

7.1 Main findings

This paper set out to test the assumption behind the 2026 reform: that SMA signatories draw more heavily on unemployment insurance because they chose to leave. Table 11 summarises the results.

On the worker side, the 2008 wager appears to have been kept: the additional cost is limited to the first two years, and it finances transitions rather than longer spells of inactivity.

Table 11: Main findings

Question	Finding	Key figures
Do signatories return more slowly to wage employment?	Slightly, and only temporarily. Most of the raw gap reflects who signs an SMA, not the SMA itself.	−2.4 pp at most; gap closed after ≈ 30 months (Section 5.1)
What do they do instead?	They start a business, train earlier and take fewer short jobs; from the third year, they hold more durable jobs.	Self-employment +5 to 6 pp; durable jobs +0.5 pp at 36 months (Section 5.2)
Is the gap a general delay?	No. Most signatories follow the same path as their controls; the difference lies in a shift towards self-employment, chosen by workers with good wage employment prospects.	Self-employment class: 21% vs 15%; same predicted wage employment as the fast-return class (Section 6)
Do they cost the UI system more?	Slightly in the first two years, then less: controls more often follow unstable paths with repeated benefit receipt.	+5.8 days at 24 months; −10.7 days at 72 months (Section 5.3.4)

7.2 The premises of the 2026 reform in workers' own words

The results above suggest that the 2008 wager has been kept on the worker side: the extra cost is limited to the first two years and finances transitions. The 2026 reform reads the same facts differently. Its justification rests on three premises that administrative data cannot test directly:

1. **The SMA is a "choice":** signatories decided to leave.
2. **They would have left anyway:** unemployment insurance subsidises a voluntary exit.
3. **The time on benefits is not productive:** shortening it costs little.

This section confronts these premises with what workers write when they negotiate, consider or have signed an SMA. Table 12 summarises the data and method; Appendix A gives the details.

Table 12: Forum analysis: data and method

Corpus	1,696 opening posts by workers who negotiate, consider or have signed an SMA, 2019–2026
Sources	Four French-speaking Reddit communities (r/conseiljuridique, r/france, r/AskFrance, r/vosfinances) and two legal advice forums (forum-juridique.net, Juritravail)
Method	Topic model by non-negative matrix factorisation: ten recurring topics, identified without imposing categories in advance
Quotes	Drawn from the posts that load most heavily on each topic

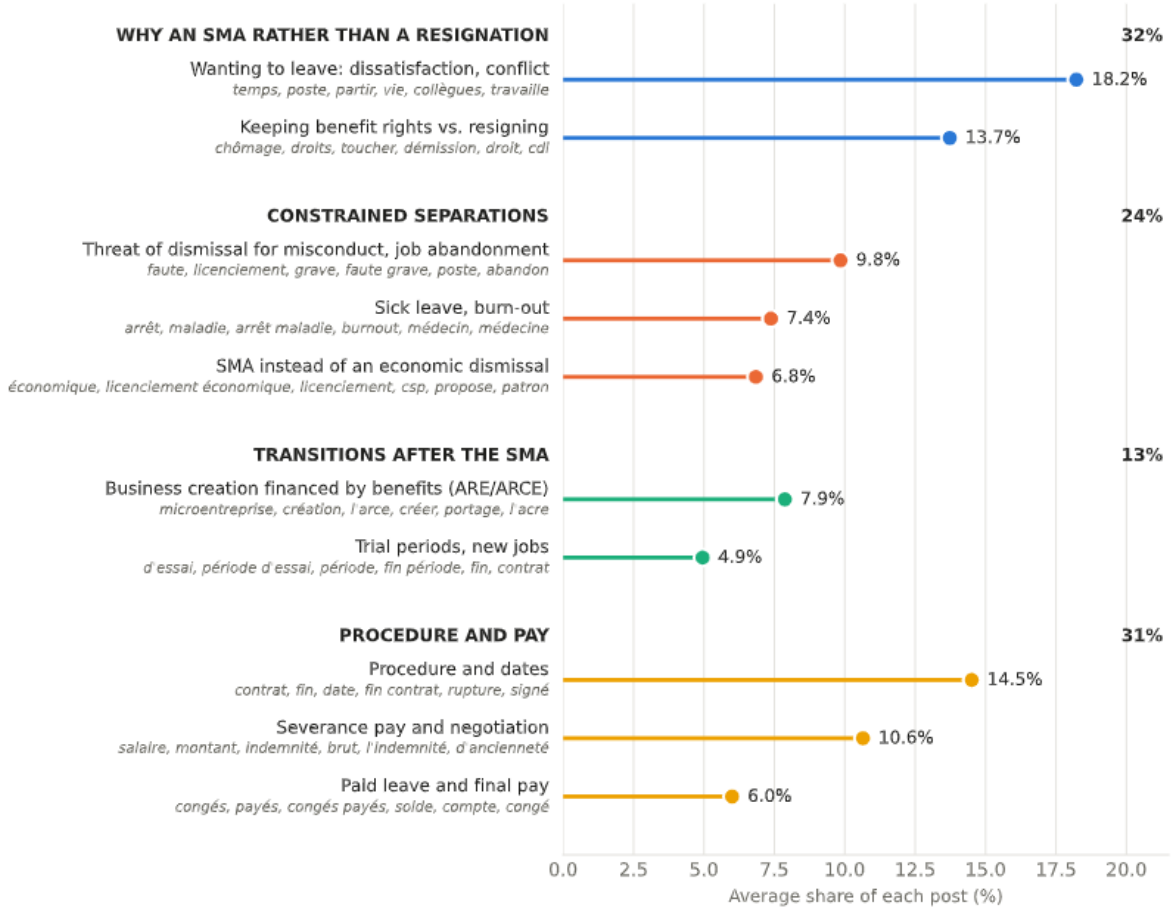


Figure 22: What workers write about the SMA: topics in online forums

Notes: Average share of each topic in 1,696 opening posts, estimated by non-negative matrix factorisation with ten topics. Words in italics are the most characteristic words of each topic (in French). Topic labels and groups are the author's. Shares describe the content of the posts, not the population of signatories. *Source:* Reddit (via Arctic Shift), forum-juridique.net and Juritravail, 2019–2026; author's calculations.

Premise 1: the SMA is a choice. For a large part of the corpus, it is. The two largest topics gather workers who want to leave their job and who seek an SMA to do so. But about a quarter of the content describes separations the worker did not seek: an SMA proposed instead of an economic dismissal ("he wants an SMA because a dismissal would cost him too much"), threats of dismissal for serious misconduct if the worker refuses to sign, SMAs negotiated during sick leave or after a burn-out. These accounts are in line with the 29% of signatories who felt compelled to leave (Bourieau, 2013) and the 24% of SMAs that replace a dismissal (Carry et al., 2025). The reform's premise holds for many signatories but not for all, and the duration cut applies to both groups alike.

Premise 2: signatories would have left anyway. The forums qualify what "anyway" means. The second largest topic is a comparison between the SMA and a resignation, which gives no entitlement: "if I resign I know I will not be entitled to benefits". When the employer refuses, posters weigh job abandonment or provoking a dismissal for misconduct, or stay. For these workers, the alternative to the SMA is not a voluntary exit financed by insurance but

an uninsured exit, or no exit at all. This is precisely the mobility the 2008 reform intended to secure. It also clarifies what the estimates of this paper measure: the matched controls left essentially through a dismissal, so the estimates compare the SMA with a dismissal. The counterfactual that matters for these workers, an uninsured resignation, is not observed in MiDAS, since resignations rarely open benefit rights.

Premise 3: the time on benefits is not productive. The business creation topic shows how part of this time is used. The project usually predates the SMA: posters already run a micro-business alongside their job or have their first clients. These business projects are rarely a way out of a health problem. In the posts centred on business creation, health (sick leave, burnout) accounts on average for less than 1% of the content, against 7% in the corpus as a whole, and this gap holds whatever threshold is used to define a business creation post. Like the predicted wage employment of the self-employment class (Section 6.2.3), this points to a planned choice rather than a fallback. The SMA is sought to keep benefit rights, and the discussion turns to how to use them, either as a monthly safety net during the first year or as a lump sum (ARCE). This matches the immediate rise in non-salaried activity at $k = 1$ (Section 5.2) and the "non-salaried, indemnified" state that opens the self-employment path (Section 6). Both instruments these posters rely on are sized by the remaining entitlement: the safety net lasts as long as the rights do, and the ARCE is a share of the rights left at the time of creation.

7.3 Was the 2026 reform well targeted?

The 2026 reform raises the employer contribution on SMA severance pay (from 30% to 40%) and cuts the maximum benefit duration for SMA claimants under 55 (15 months, against 18 after any other separation). Table 13 confronts the second measure with the results.

Table 13: The duration cut in light of the results

Reform logic	What the results show	Implication
Signatories stay longer on benefits	True only in the first two years, and slightly; over six years they consume fewer benefits than comparable workers.	The cut targets a cost that is small and temporary once selection is accounted for.
Shorter benefits speed up the return to work	Employment converges when benefits run out, but this time is used to start a business, train or find a durable job.	The cut may reduce short-run costs while weakening the transitions the SMA finances.
All SMA claimants are concerned	Differences are concentrated in a minority of paths; most signatories behave like their controls.	The cut mainly affects those who use their entitlement to build a transition.
The SMA increases UI spending	Not addressed here: the paper compares workers who all claimed benefits. Only about a quarter of SMAs replace a dismissal (Carry et al., 2025).	If the cost lies in entry into the scheme, the employer contribution is better targeted than a duration cut.

The duration cut therefore acts on the behaviour of signatories after the separation, where the

results show little extra cost, rather than on the decision to separate, where the cost of the scheme is more likely to lie.

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A Online forums: corpus, method and robustness

Corpus. The corpus gathers opening posts, published between 2019 and 2026, in which the author discusses an SMA they are negotiating, considering or have signed. Reddit posts (r/conseiljuridique: 1,203; r/vosfinances: 154; r/france: 132; r/AskFrance: 81) were collected through the Arctic Shift archive, and 126 posts come from forum-juridique.net and Juritravail. Replies, duplicates, posts in English and posts shorter than 150 characters are excluded, leaving 1,696 posts. User names are removed.

Pre-processing. Posts are lower-cased; URLs and numbers are removed, spelling variants are merged (*rupture co*, *RC*; *Pôle emploi*, *France Travail*), and French stop words are dropped. The vocabulary contains single words and pairs of words (*arrêt maladie*) that appear in at least 8 posts and at most 35% of them: $V = 3{,}288$ terms for $D = 1{,}696$ posts.

TF-IDF and NMF. Counts are weighted by term frequency–inverse document frequency, which down-weights terms that are common to all posts. For term t in post d,

$$x_{dt} = (1 + \ln n_{dt}) \left(\ln \frac{1 + D}{1 + D_t} + 1 \right), \tag{8}$$

where n_{dt} is the number of occurrences of t in d and D_t the number of posts containing t; each row is then normalised to unit length. The topic model is a non-negative matrix factorisation (Lee and Seung, 1999) of the resulting matrix X:

$$X \approx WH, \qquad (\hat{W}, \hat{H}) = \arg \min_{W,\, H \geq 0} \|X - WH\|_F^2, \tag{9}$$

where H $(K \times V)$ gives the weight of each term in each of the K topics and W $(D \times K)$ the weight of each topic in each post. Non-negativity makes each post an additive mixture of topics. The share of topic k in post d is $\theta_{dk} = \hat{w}_{dk} / \sum_j \hat{w}_{dj}$, and Figure 22 reports its average across posts. Topic labels were assigned after reading the posts with the largest share of each topic.

Number of topics. Two criteria are used (Table 14). Coherence checks that the ten most characteristic terms of a topic appear together in the posts, using the normalised pointwise mutual information

$$\mathrm{NPMI}(a, b) = \ln \frac{P(a, b)}{P(a)\, P(b)} \Big/ \big(- \ln P(a, b) \big), \tag{10}$$

where $P(a)$ and $P(a, b)$ are the shares of posts containing a, and both a and b. Stability checks that the same topics are found from six random starting points, by matching topics across runs on the cosine similarity of their rows of H. Ten is the only number of topics for which every topic is recovered in every run; with fewer or more, a small topic splits or merges from one run to the next. Latent Dirichlet allocation, estimated for comparison, yields markedly less coherent topics at every K.

Table 14: Choice of the number of topics

	Coherence (NPMI)		Stable topics
K	NMF	LDA	(similarity ≥ 0.9)

B Glossary

Table 15: Glossary

Acronym	**Meaning**
AAH	*Allocation aux adultes handicapés*, disability allowance
ARCE	*Aide à la reprise ou à la création d'entreprise*, capital payment of remaining ARE rights to business creators
ARE	*Allocation d'aide au retour à l'emploi*, unemployment insurance benefit
ATT	Average treatment effect on the treated
CDD / CDI	Fixed-term contract / open-ended contract
CNAF	National family allowance fund
DARES	Research and statistics department of the Ministry of Labour
DSN	*Déclaration sociale nominative*, monthly employer payroll declaration
LFSS	Social security financing law
MiDAS	*Minima sociaux, droits d'assurance chômage et parcours salariés*
MMO	*Mouvements de main-d'œuvre*, labour movements file
RCC / SMA	Collective / separation by mutual agreement
RSA	*Revenu de solidarité active*, minimum income
SMD	Standardised mean difference